

Date

[Micro Economics]

Positive - How things are

Normative - How things should be

Opportunity cost - Second best alternative

5 kg Rice or
10 kg wheat

$$OC_x = \frac{\text{Loss in } X}{\text{Gain in } X} = \frac{\text{Loss in wheat}}{\text{Gain in Rice}}$$

$$= \frac{10}{5} = 2$$

Intuition: For every kg of rice I grow, I have to forgo 2 kg of wheat

Implicit pricing -

Pizza > cake

cake > tea

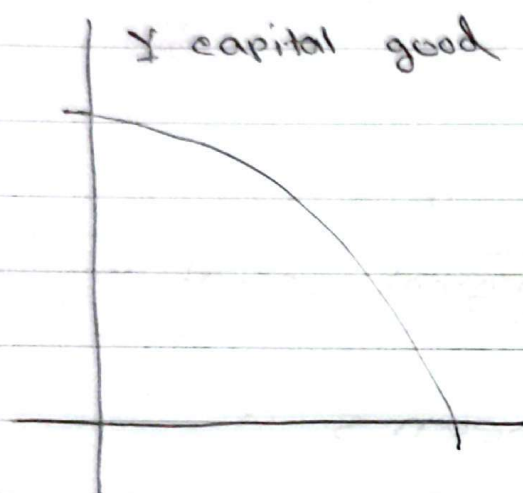
So, Pizza > Tea

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Production possibilities comparative/ absolute advantage and trade

Production possibilities curve/

Frontier (PPC/ PPF) : A curve showing the maximum combinations of goods

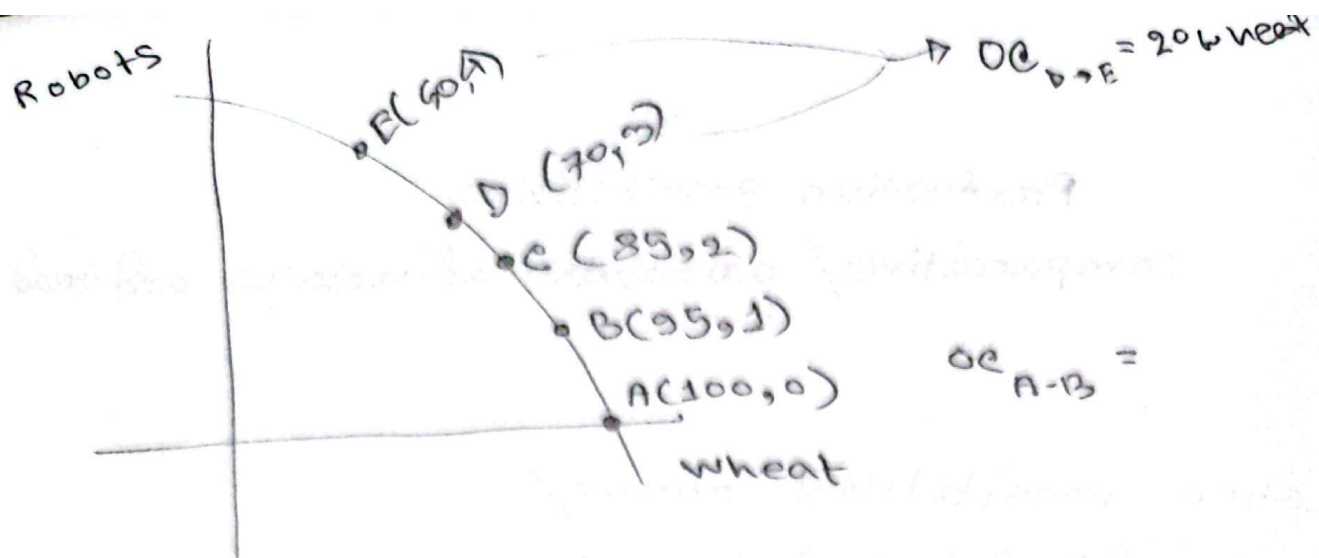


Assumptions:

- 1) Fixed resource
- 2) Fixed technology
- 3) Two goods economy

Law of increasing opportunity cost:

↳ As you produce more you have to give up increasing amounts of the other good.

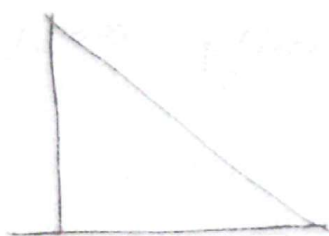


constant opportunity cost : $Y = mx + b$

marginal opportunity cost

A 'steeper' tangent means a greater change per unit

↳ greater opportunity cost



high
oc



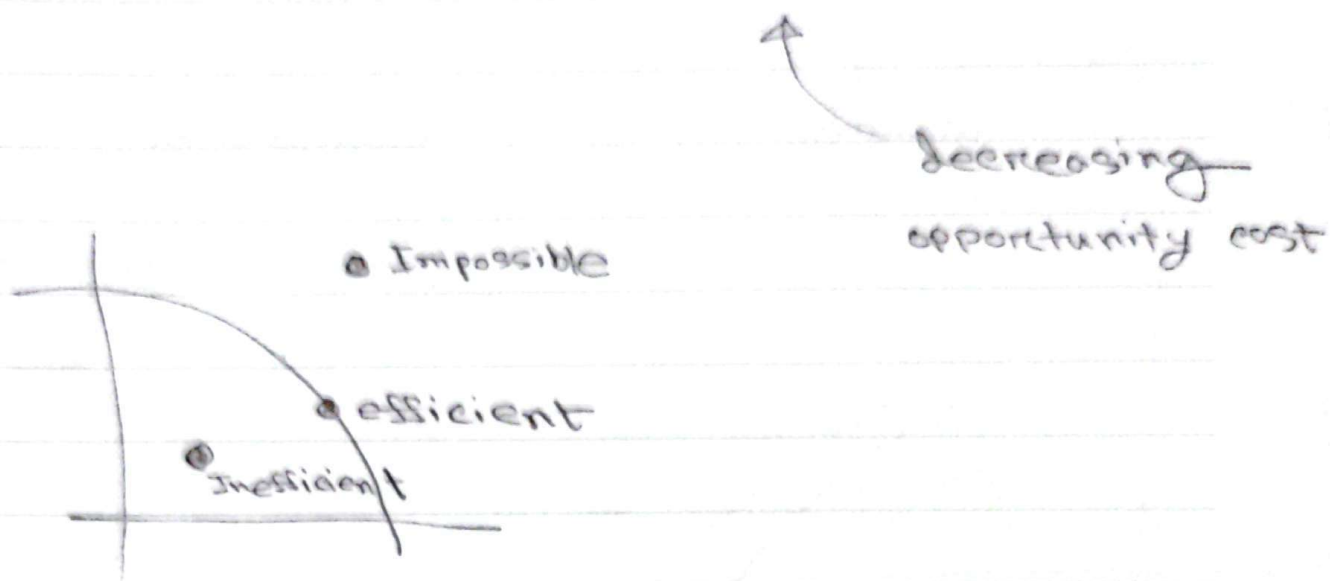
low
oc



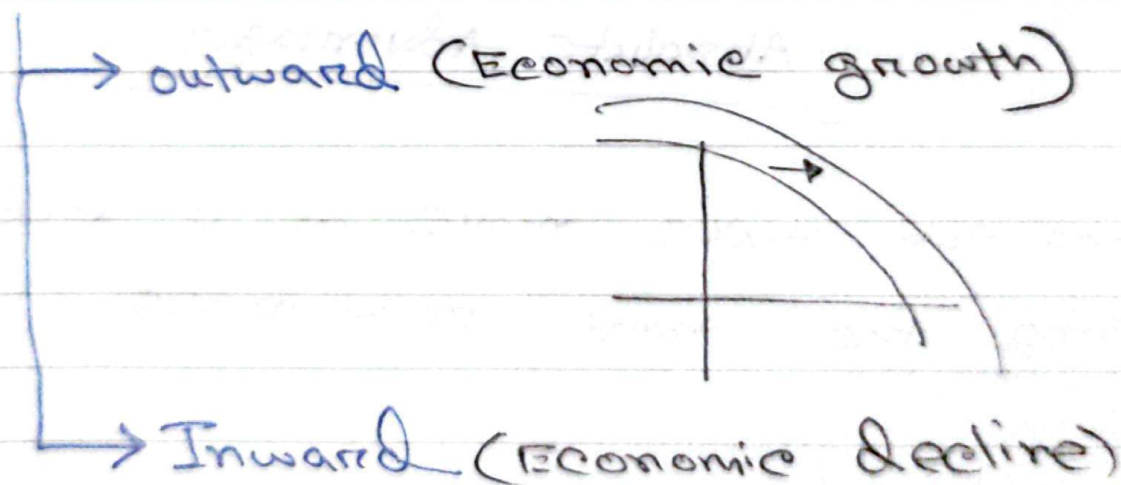
zero
oc

Date

PPX convex and not concave:



Shifts



- Natural disasters
- war
- Epidemics
- Institutional collapse

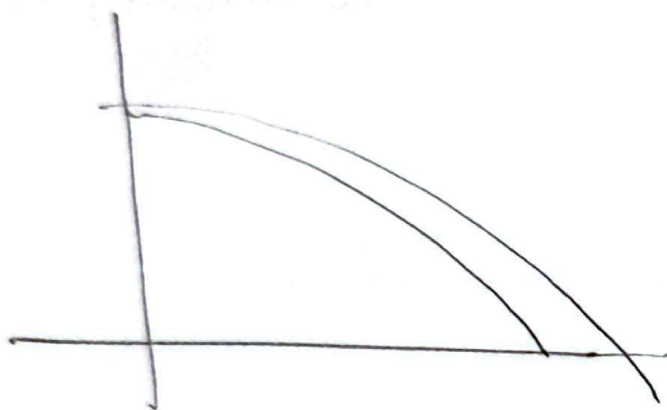
RANGS PHARMA

- Resource depletion

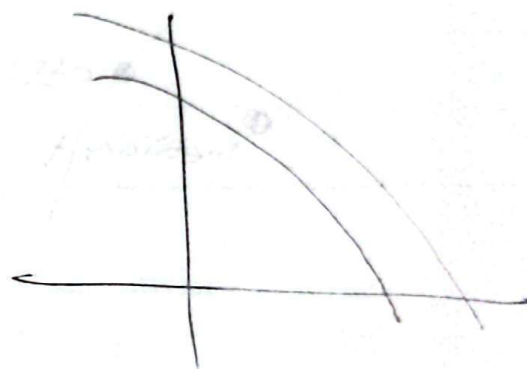
Types of PPF

Biased

Unbiased



(Parallel growth)



Absolute Advantage

who can make more of a good using the same resources.

BD - 10 cars - 5 kg steel

JP - 20 cars - 12 kg "

BD productivity = $\frac{10}{5}$ cars per kg = 2 cars / kg

JP " = $\frac{20}{12}$ " " = 1.67 "

BD has the absolute advantage.

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Comparative Advantage

It's about who gives up less

Is I lose less $\rightarrow$ Oc less

$$\text{Term of trade} = \frac{\text{units of good exported}}{\text{units of good imported}}$$

If oc are equal

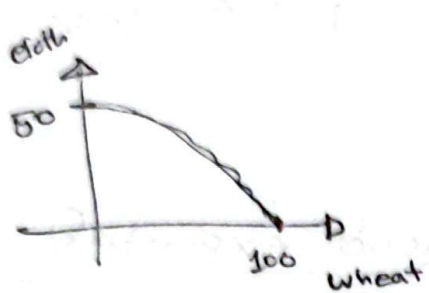
$\rightarrow$ no trade occurs

①

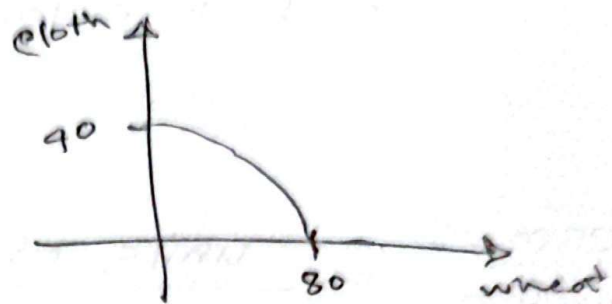
A — $\begin{cases} \rightarrow 100 & \text{w/day} \\ \rightarrow 50 & \text{c/day} \end{cases}$
B — $\begin{cases} \rightarrow 80 & \text{w/day} \\ \rightarrow 40 & \text{c/day} \end{cases}$

a)

PPF:



Ⓐ



Ⓑ

b)

Absolute adv.,

A $\frac{\text{wheat}}{\text{cloth}} = \frac{100}{50} = 2$ A $\frac{\text{cloth}}{\text{wheat}} = \frac{50}{100} = 0.5$
B $\frac{\text{wheat}}{\text{cloth}} = \frac{80}{40} = 2$ B $\frac{\text{cloth}}{\text{wheat}} = \frac{40}{80} = 0.5$

	wheat	oc	cloth	oc
A	100	0.5	50	2
B	80	0.5	40	2

oc_{wheat} A = $\frac{50}{100} = 0.5$

oc_{wheat} B = $\frac{40}{80} = 0.5$

oc_{cloth} A = $\frac{100}{50} = 2$

oc_{cloth} B = $\frac{80}{40} = 2$

c)

d) oc equal, neither has comparative adv.

e) oc equal, so, trade may not be beneficial in this case,

Date

② $y = mx + c$
 $\Rightarrow y = -3x + c$

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x}$$
$$\Rightarrow -3 = \frac{\Delta y}{\Delta x}$$

$$\Rightarrow -3 \Delta x = \Delta y$$

$\therefore$ 1 unit of $X = 3$ units of Y

So, $OC_X = 3Y$
 $OC_Y = \frac{1}{3X}$

③

$$C = 100 - 2W$$

$$y = +c - mx$$

$$C = -2W + 100$$

$\uparrow$
 m

opportunity cost of 1 car,

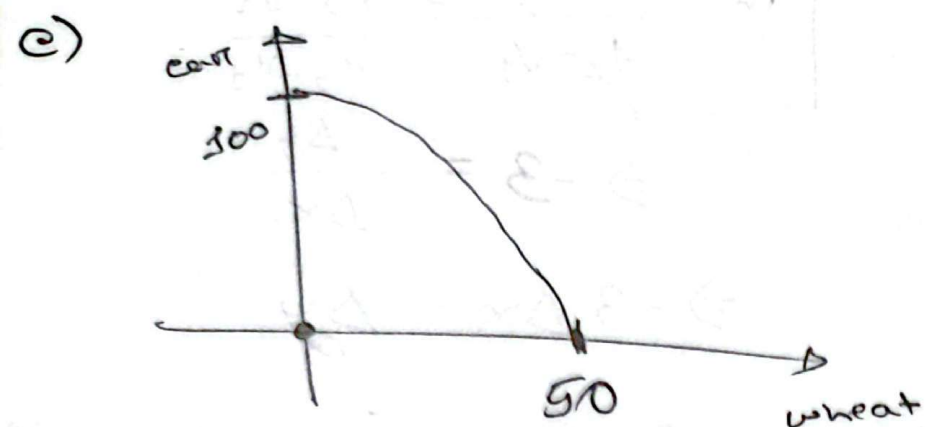
~~$1 = 100 - 2W$
 $\Rightarrow 2W = 99$
 $\Rightarrow W = \frac{99}{2}$~~

a) $OC_C = \frac{1}{m}$

$$= \frac{1}{2}$$

= 0.5 wheat

b) $OE_{\text{wheat}} = 2 \text{ cars}$



$$C = 100 - 2 \times 0$$

$$= 100$$

$$\left. \begin{array}{l} x = 0 \\ y = 100 \end{array} \right\}$$

$$C = 100 - 2 \times 50$$

$$= 0$$

$$\left. \begin{array}{l} x = 50 \\ y = 0 \end{array} \right\}$$

Date

		Cars	oc _{ca}	Computers	oc _{co}
④	A	20	2	40	0.5
a)	B	15	3	45	0.3

Cars → A

Comp. → B

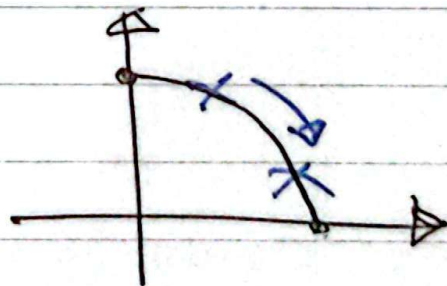
b) B → adv. in comp.
comparative

c) A → Cars

B → computers

d) Trade 1 car for 2 computers

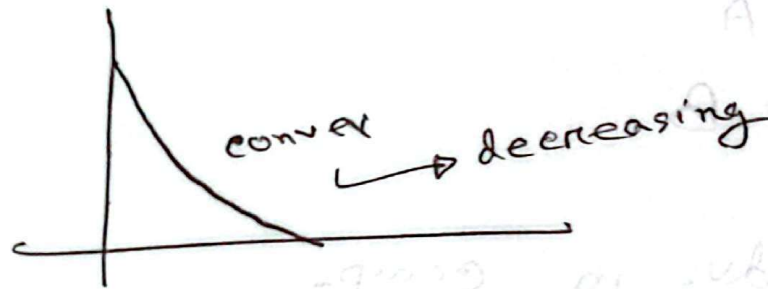
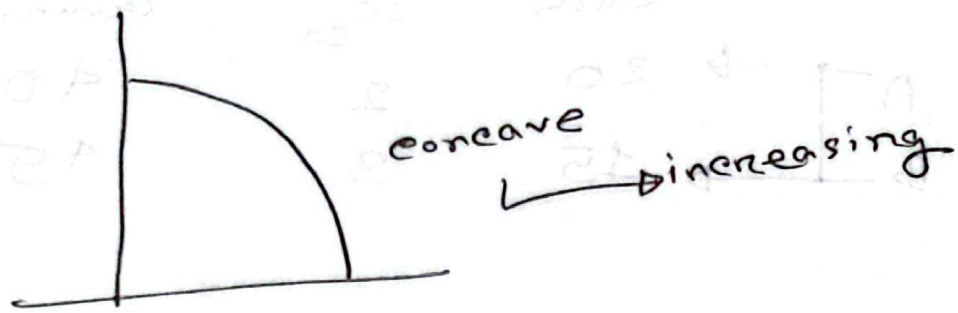
⑤ Inconsistent
efficient



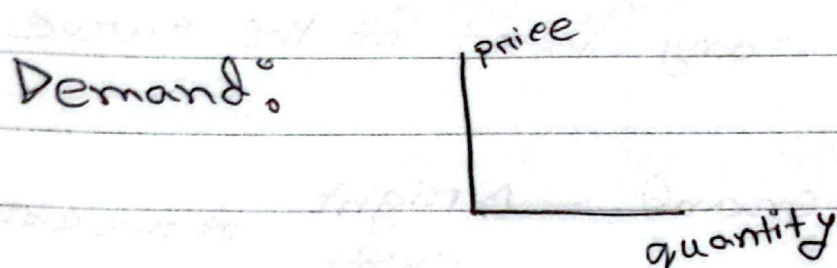
It represents gain in efficiency
-the country is now using all of
its resources.

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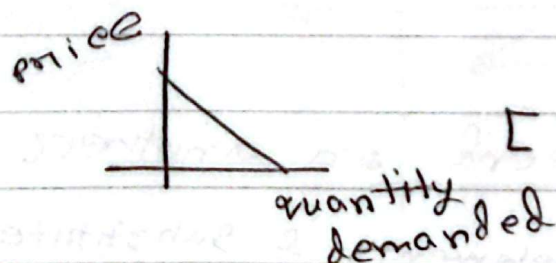


Both works together

Demand is how much good & or service people want to buy at a given price, when they are both willing and able to pay for it.

[Demand + capability] $\rightarrow$ need &

[more less price $\rightarrow$ more interested people to buy]



[downward curve;
negative slope]

① Substitution
Effect

$\rightarrow$ If the price of tea is increased people will move to other drinks like coffee

② Income Effect $\rightarrow$

③ Diminishing marginal utility $\rightarrow$ buying item for the second time, the rate of change for the next unit will be ~~more~~ less interesting.

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quantity demand: - any point on the curve

more demand $\rightarrow$ right shift of the curve

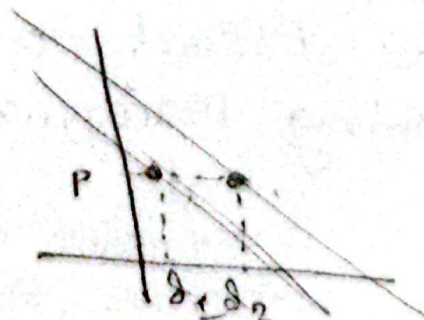
less demand $\rightarrow$ left shift " " "

inferior goods
 $\rightarrow$ more income but less demand

normal goods
 $\rightarrow$ more income and more demand

✓ 1 product can depend on another product too. (complements & substitutes)

while shifting, the prices will be the same, quantity demand will change.



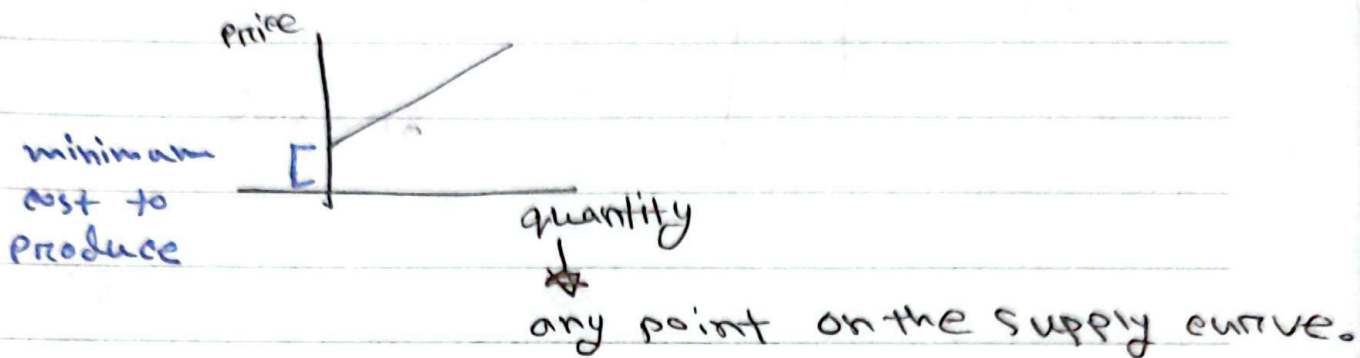
[we only think about in consumers here]

Date

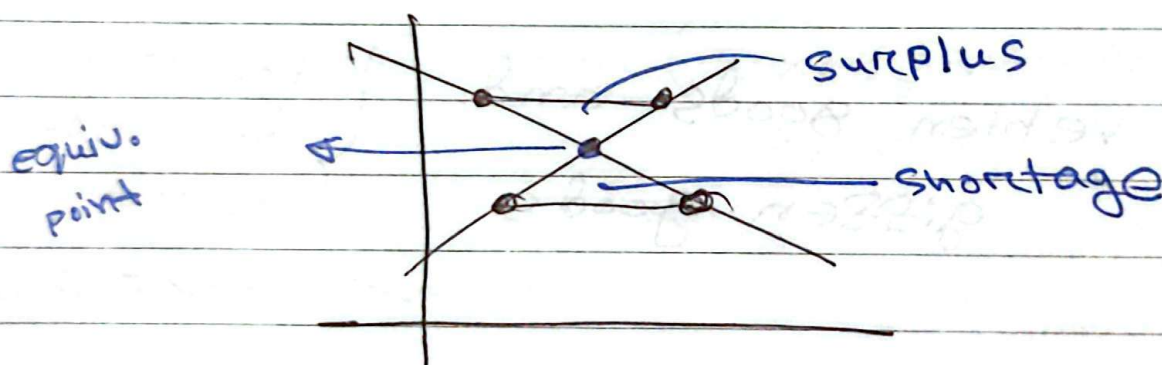
opposite direction of method

Supply :

Price goes up $\rightarrow$ supply goes up



equivalent price $\rightarrow$ supply demand same



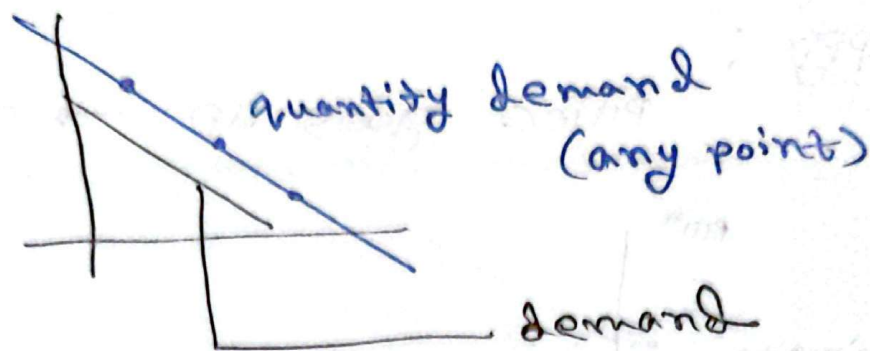
$$Q_d = 10 + 5P$$

$$Q_s = 20 + 7P$$

$$Q_d = Q_s \rightarrow \text{equiv. point}$$

[we only think about
producers in here]

RANGS PHARMA

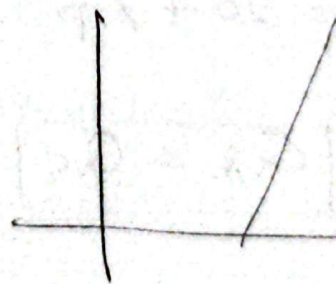
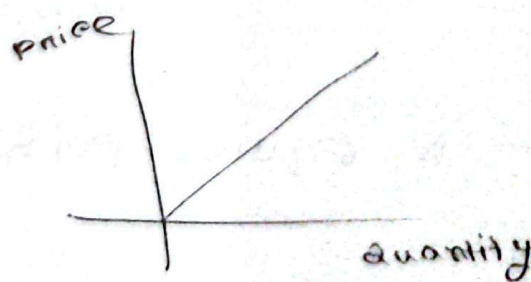


Demand equation: $Q_d = 1000 - 2P + 100$

veblen goods and
giffen goods

the law
demand
is not
universal

more price = more demand



Date

more price $\rightarrow$ supplier want to produce more because of more profit.

why does supply shift -

1) price of Inputs/resources

$\rightarrow$ steel for the car

2) Technology

3) Govt. Action (Taxes, subsidies, regulation)

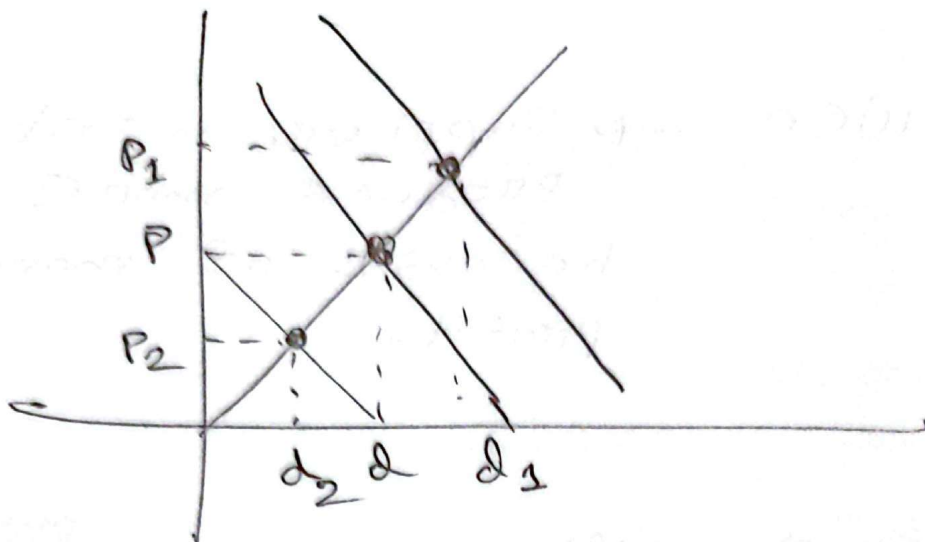
4) Number of sellers

5) Expectations

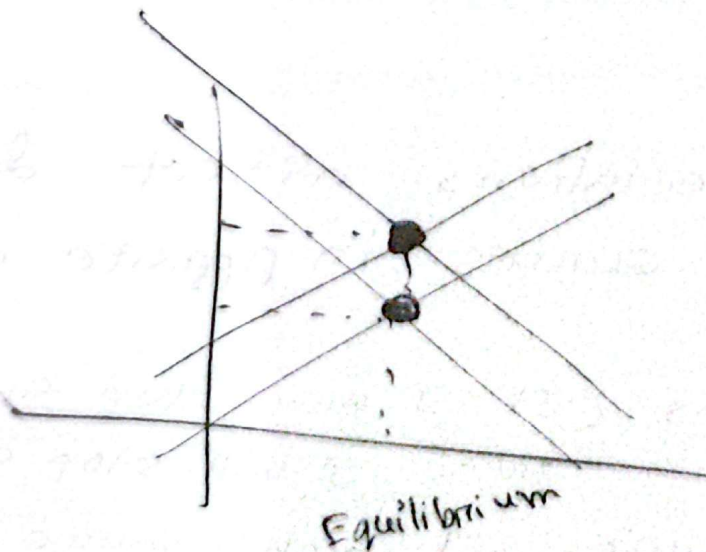
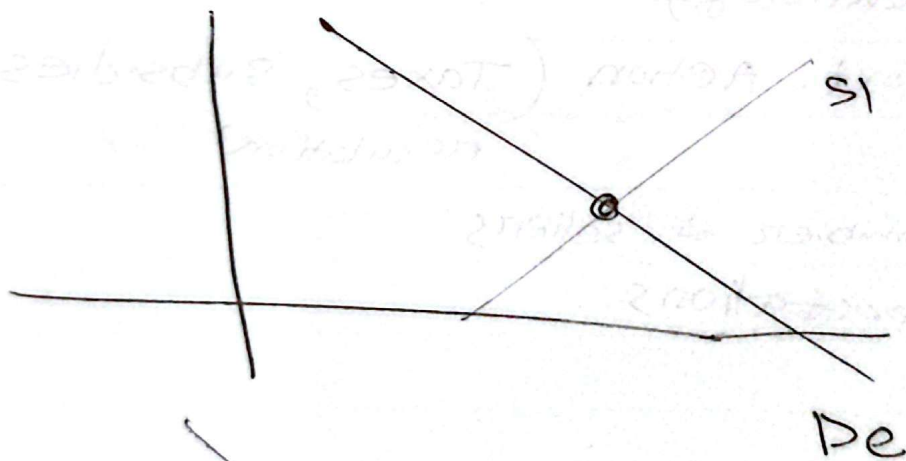
Price & expectations affect demand and supply curves in opposite ways.

Ex- Lpg gas (If I know the future price, I will stop supply which will make more profit)

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more supply $\rightarrow$ right shift
 intersection point also changes



Date

$$Q_d = 100 - 5P \quad \text{--- (i)}$$

$$Q_s = 50 + 3P \quad \text{--- (ii)}$$

$$Q_d = Q_s$$

$$\Rightarrow 100 - 5P = 50 + 3P$$

$$\Rightarrow 8P = 50$$

$$\Rightarrow P = \frac{50}{8}$$

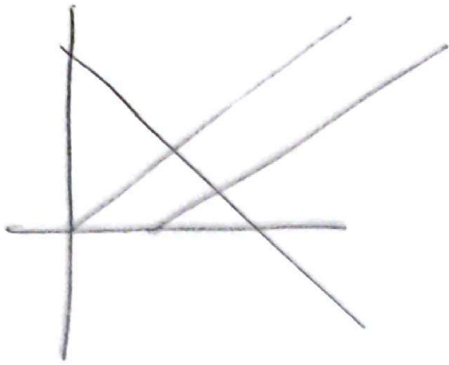
$$(i) \Rightarrow Q_d = 100 - 5 \times \frac{50}{8}$$

$$= 100 - \frac{250}{8}$$

$$= 31.2$$

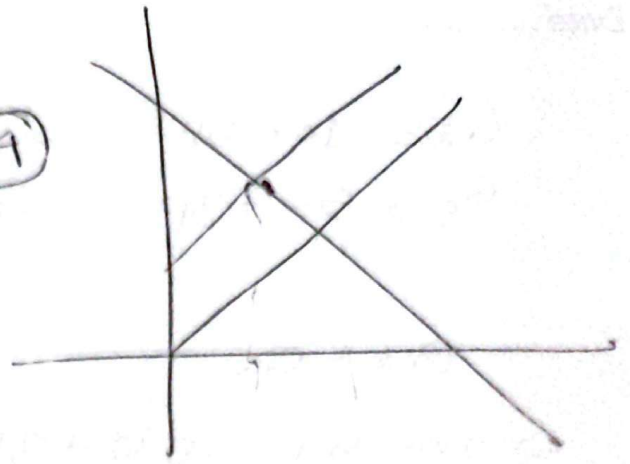
$$\begin{array}{r} 8 \overline{) 250} (31.2 \\ \underline{24} \\ 10 \\ \underline{8} \\ 20 \\ \underline{16} \\ 4 \end{array}$$

③

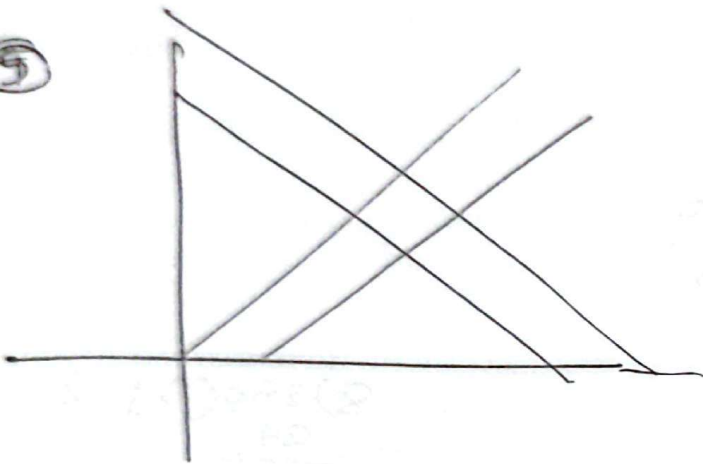


dual shift →

④

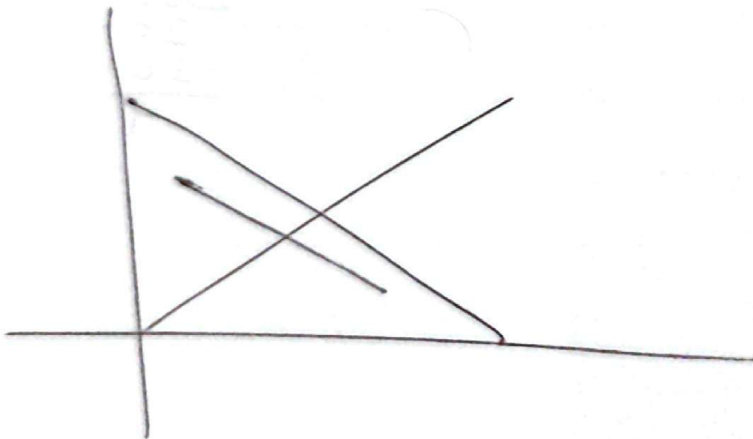


⑤

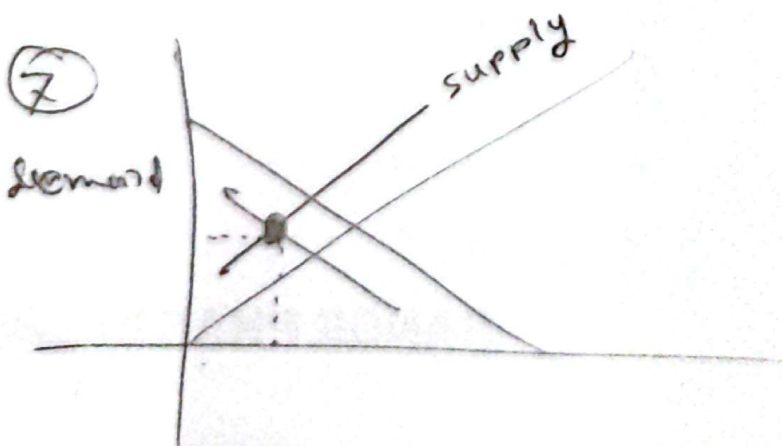


both shift right
price ambiguous

⑥



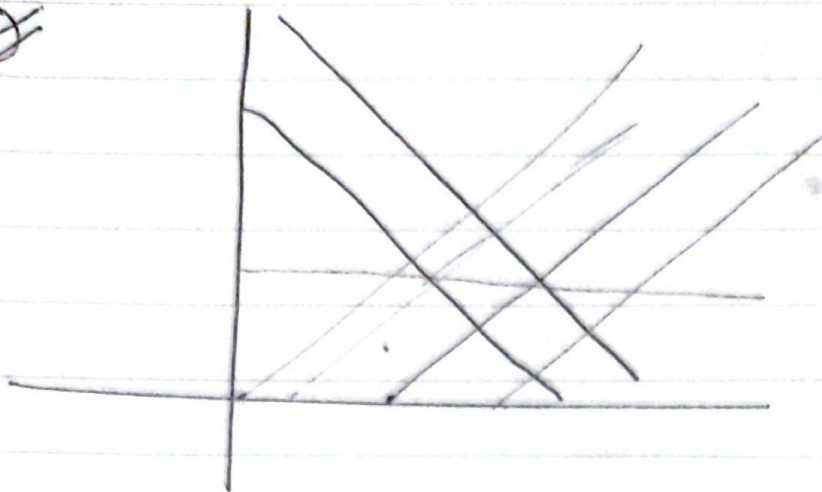
⑦



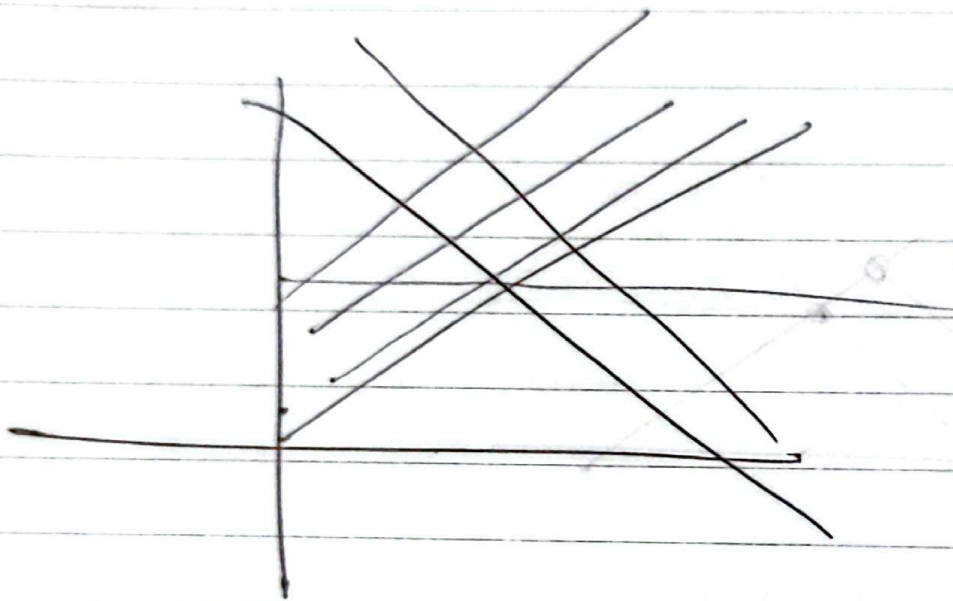
price ambiguous
↑
both shifts together

Date

8



price
ambiguous

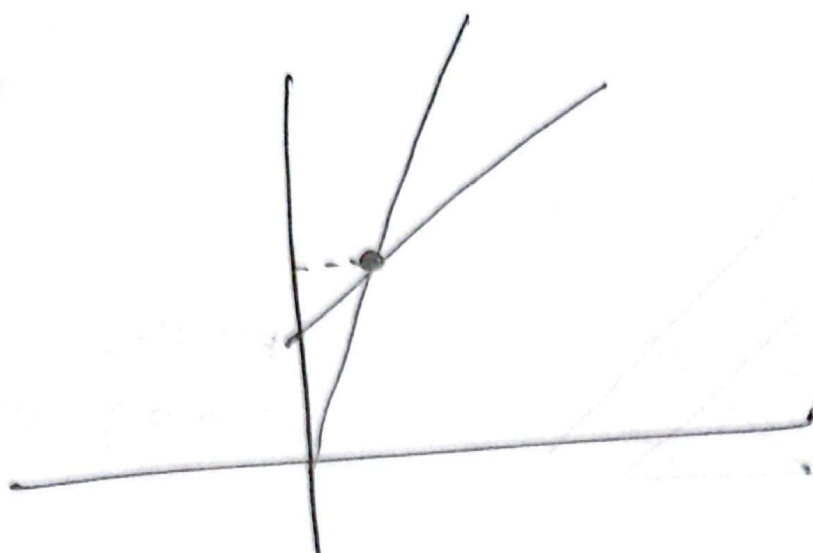


opposite shift

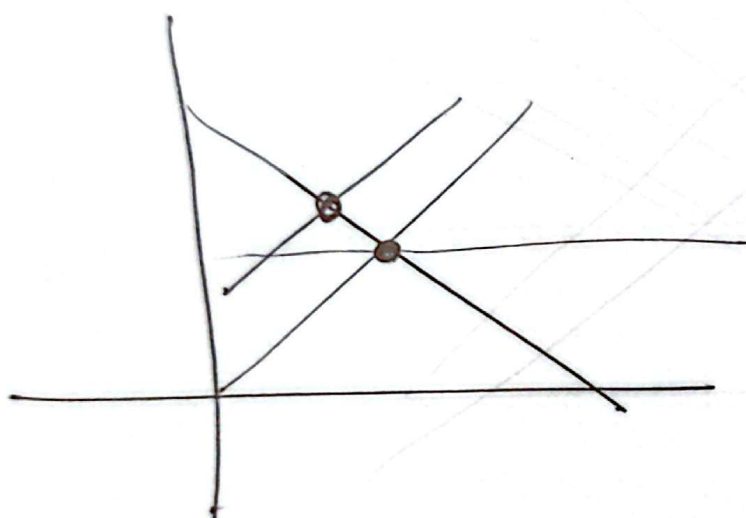


quantity ambiguous

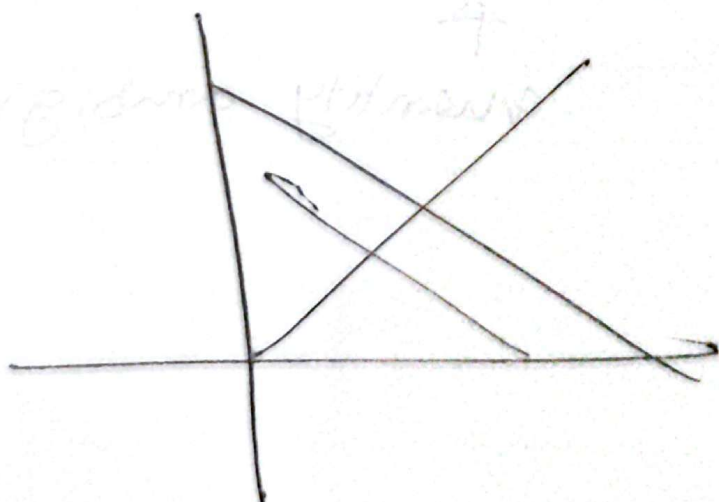
8)



9)

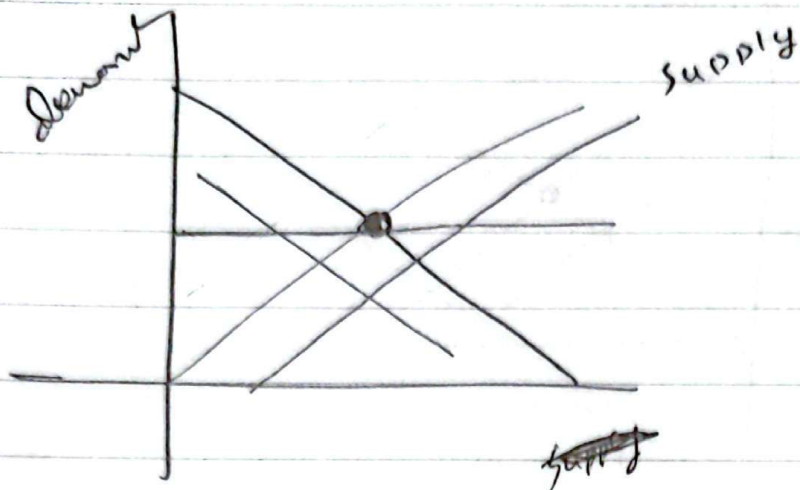


10)



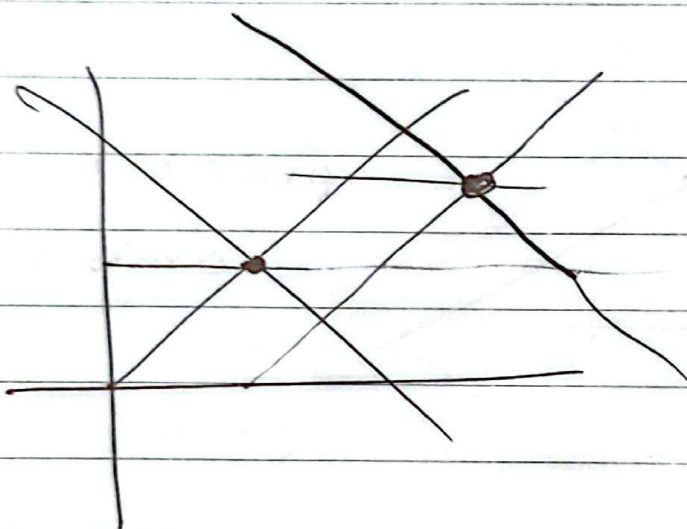
Date

11)



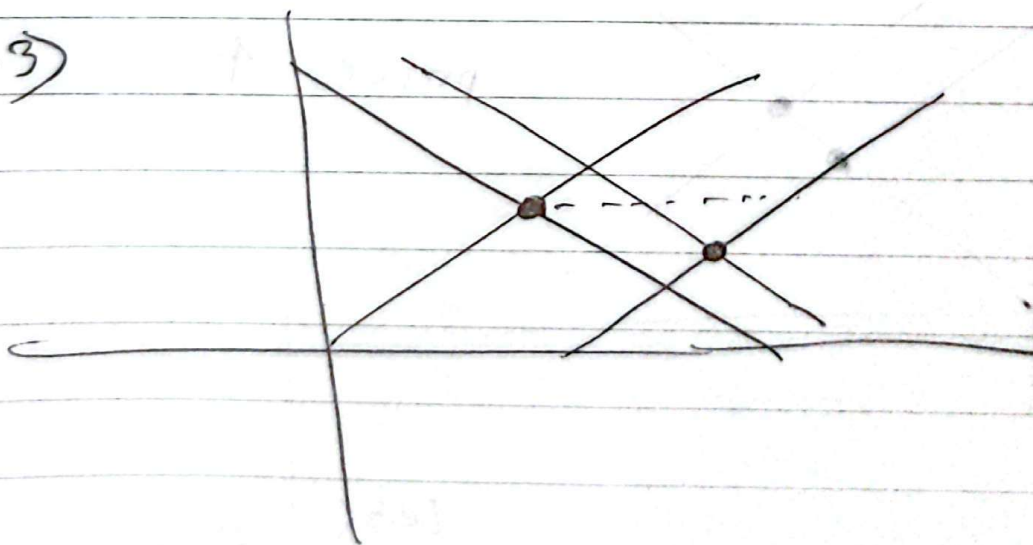
quantity
ambiguous;
price will drop

12)

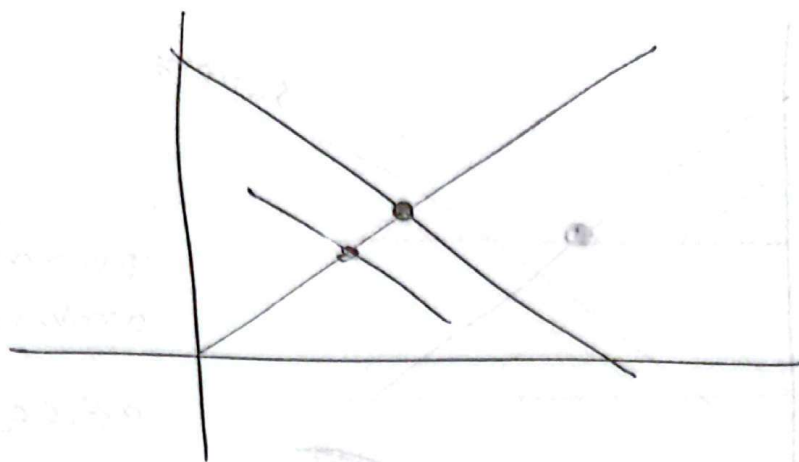


more price
more quantity

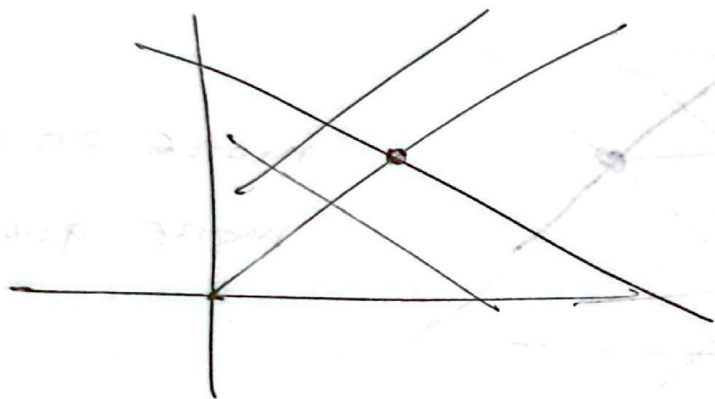
13)



14)

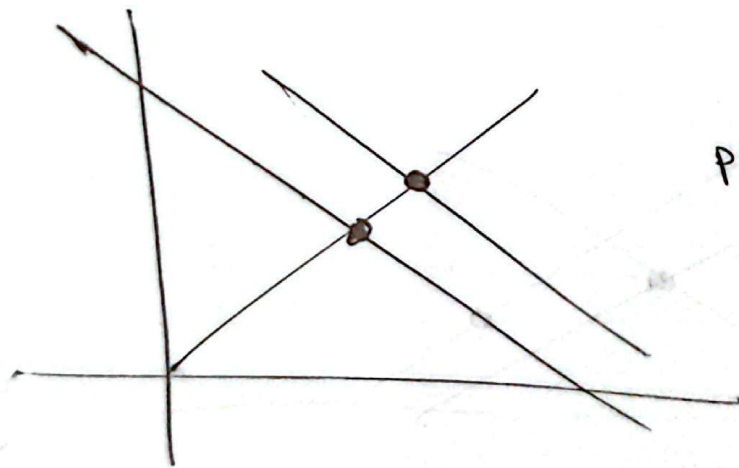


15)



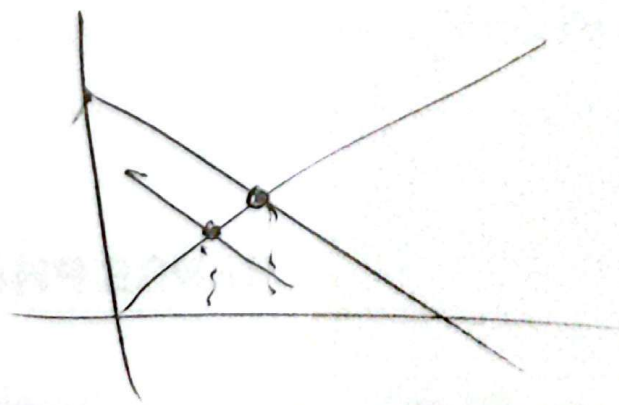
price $\rightarrow$
• ambiguous

16)



price $\uparrow$

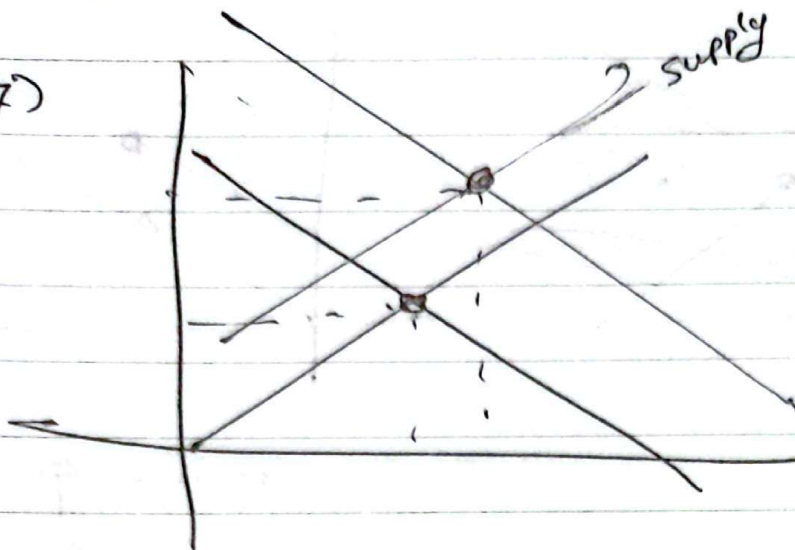
17)



less demand

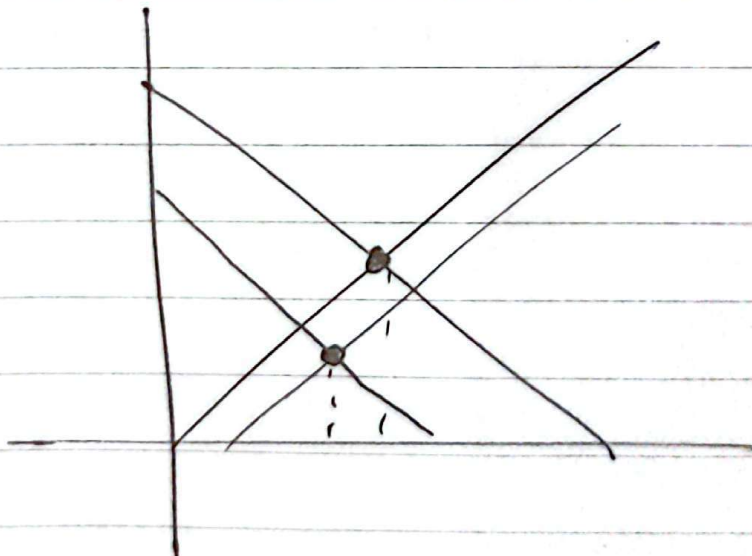
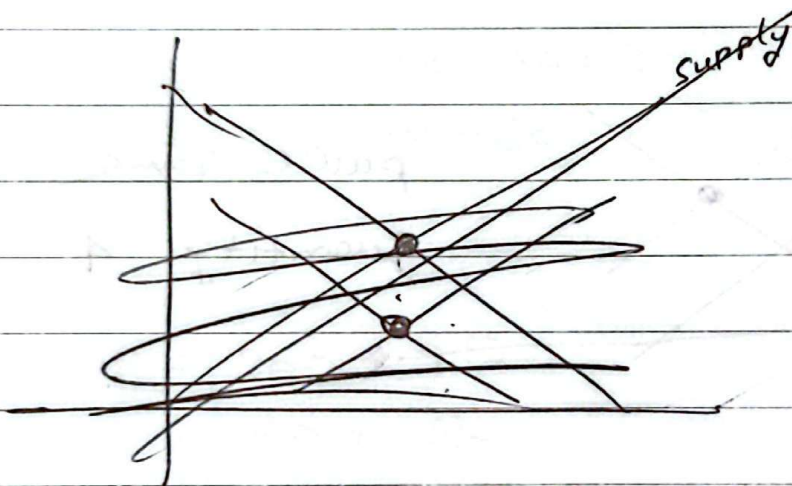
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17)

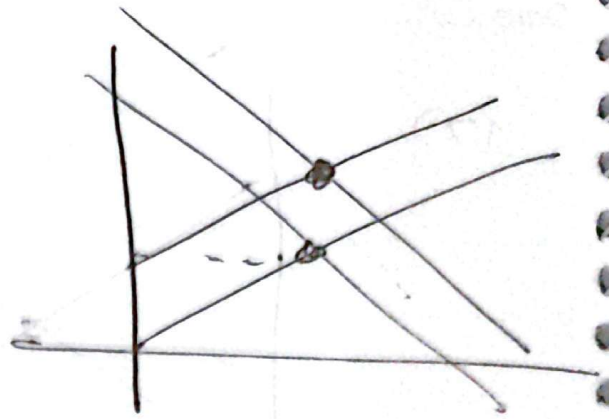
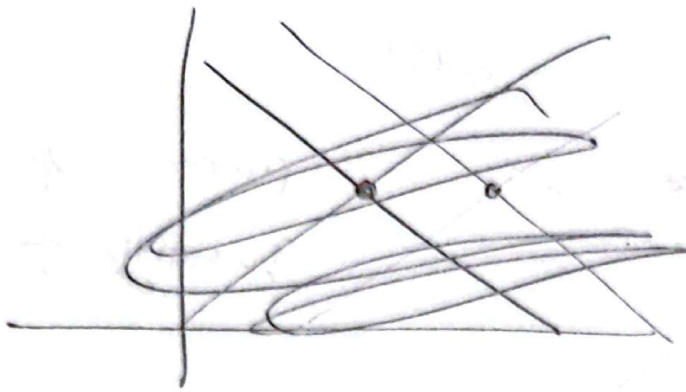


price ↑
quantity ↑

18)

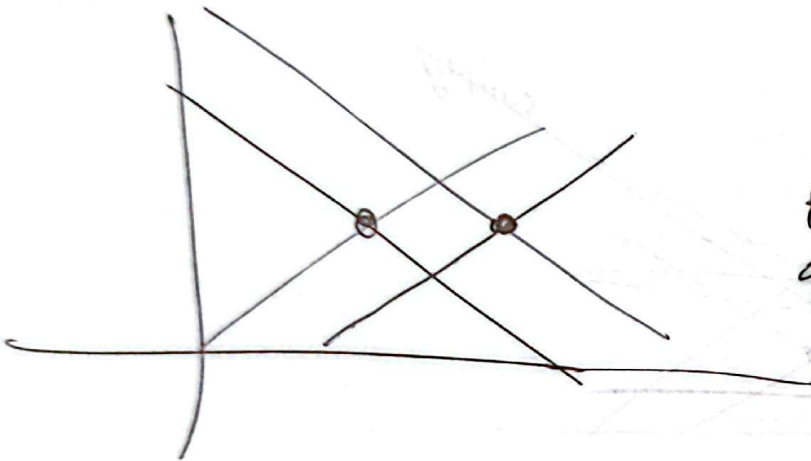


19)



price $\uparrow$
quantity same

19)



price same
quantity $\uparrow$

Date

a)

$$D: Q_D = 120 - 2P$$

$$S: Q_S = 20 + 3P$$

$$120 - 2P = 20 + 3P$$

$$\Rightarrow 5P = 100$$

$$\Rightarrow P = 20$$

$$\therefore Q_D = 120 - 40 = 80$$

$$Q_S = 20 + 60 = 80$$

$$\# \quad P = 200 - 4Q$$

$$P = 40 + Q$$

(-)

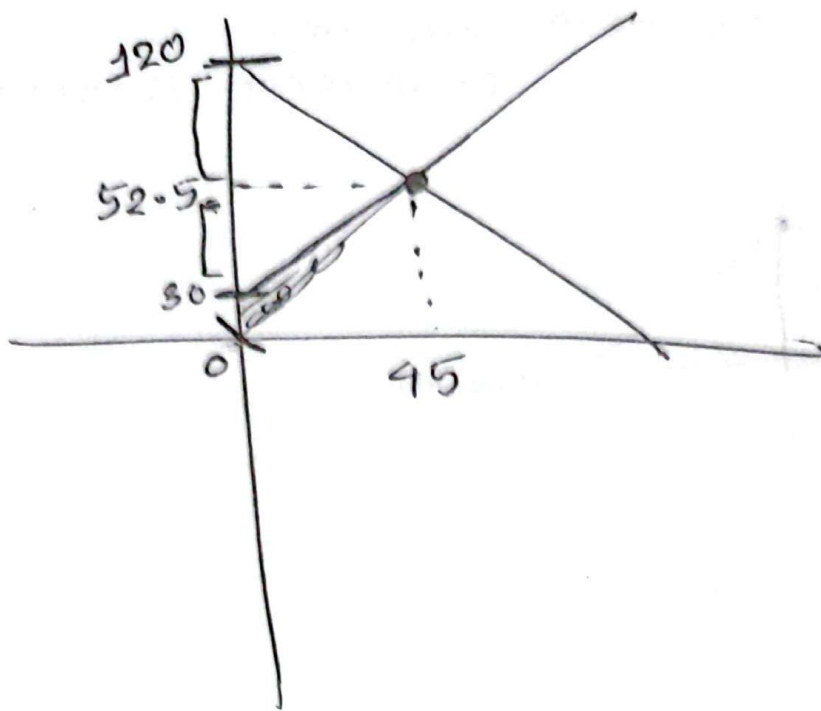
$$160 = 160 - 5Q$$

$$\Rightarrow 5Q = 160$$

$$\Rightarrow Q = \frac{160}{5} = 32$$

$$\begin{array}{r} 5) 160(32 \\ \underline{15} \\ 10 \\ \underline{10} \\ 0 \end{array}$$

$$P = 40 + 32 = 72$$



$$C_s = \frac{1}{2} \times 45 \times (120 - 52.5)$$

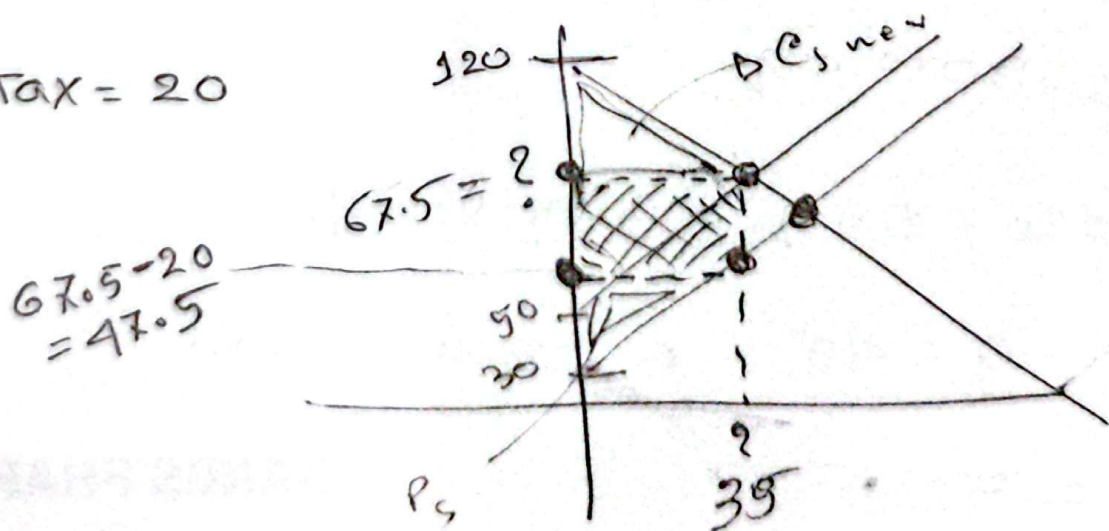
$$= 1518.75$$

$$P_s = \frac{1}{2} \times 45 \times (52.5 - 30)$$

$$= 506.25$$

b)

Tax = 20



Date

$$P_d = 120 - 1.5q$$

$$P_{S_{\text{new}}} = 30 + 0.5q + 20$$

$$P_d = P_{S_{\text{new}}}$$

$$\Rightarrow 120 - 1.5q = 30 + 0.5q + 20$$

$$\therefore q_{\text{new}} = 35$$

$$\begin{aligned}\therefore P_{d_{\text{new}}} &= 120 - 1.5 \times 35 \\ &= 67.5\end{aligned}$$

$$\begin{aligned}\text{Tax} &= L \times B \\ &= 35 \times 20 \\ &= 700\end{aligned}$$

$$\begin{aligned}C_{S_{\text{new}}} &= \frac{1}{2} \times 35 \times (120 - 67.5) \\ &= 918.75\end{aligned}$$

$$\begin{aligned}P_{S_{\text{new}}} &= \frac{1}{2} \times 35 \times (47.5 - 30) \\ &= 306.\end{aligned}$$

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$$D_{WL} = \frac{1}{2} \times \text{tax} \times (45 - 35)$$

$$= \frac{1}{2} \times 20 \times 10$$

$$= 100$$

Date

$$300 - 2 \cdot 0$$

$$\# P = 300 - 2q$$

$$P = 60 + q$$

$$P = 0, q = 0$$

P → price → 140

q → quantity → 80

$$\Rightarrow 300 - 2q = 60 + q$$

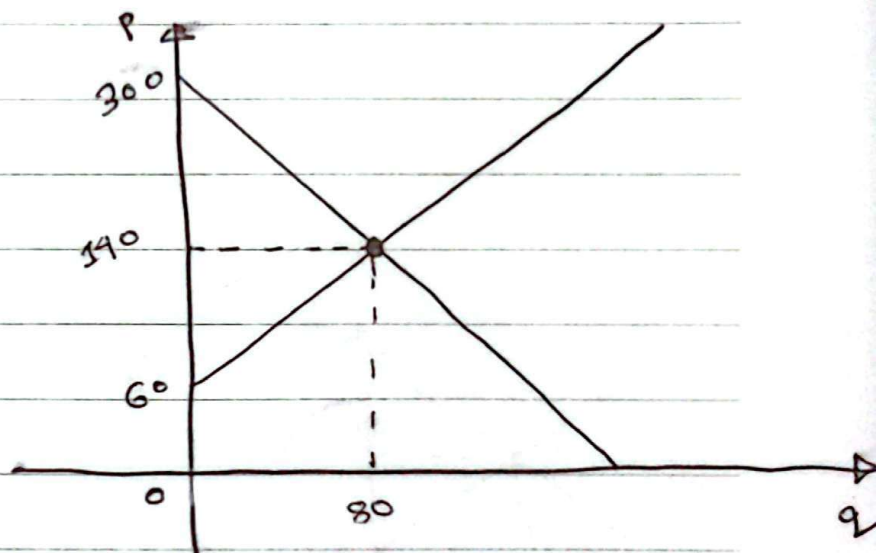
$$\Rightarrow 300 - 60 = 3q$$

$$\Rightarrow 240 = 3q$$

$$\Rightarrow q = 80$$

$$\therefore P = 60 + 80$$

$$= 140$$



$$C_s = \frac{1}{2} \times 80 \times (300 - 140)$$

$$P_s = \frac{1}{2} \times 80 \times (140 - 60)$$

$$\text{Tax} = 30$$

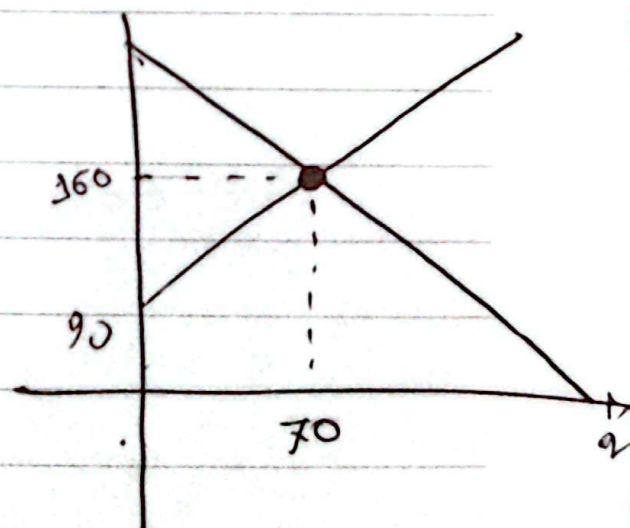
$$P_d = 300 - 2q$$

$$P_{s, \text{new}} = 60 + q + 30$$

$$\Rightarrow 300 - 2q = 60 + q + 30$$

$$\Rightarrow 210 = 3q$$

$$\Rightarrow q = 70 \quad \Rightarrow P = 160$$



$$C_s' = \frac{1}{2} \times 70 \times$$

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Date

Q-2:

$$a) P = 250 - Q$$

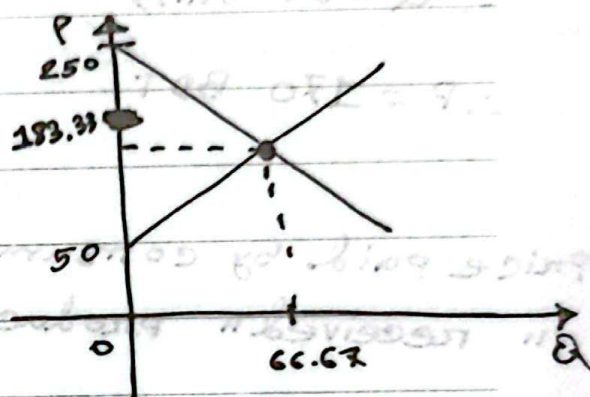
$$P = 50 + 2Q$$

$$\Rightarrow 250 - Q = 50 + 2Q$$

$$\Rightarrow 3Q = 200$$

$$\Rightarrow Q = \frac{200}{3} = 66.67 \text{ units}$$

$$\therefore P = 50 + 2 \times \frac{200}{3}$$
$$= 183.33 \text{ BDT}$$



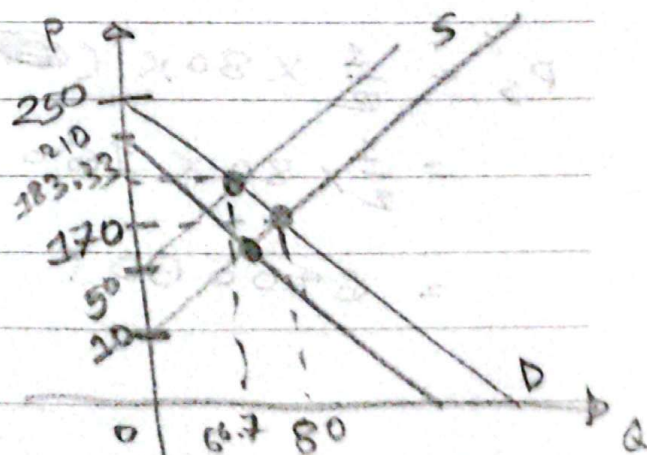
$$C_S = \frac{1}{2} \times 66.67 \times (250 - 183.33)$$

$$= \frac{1}{2} \times 66.67 \times 66.67$$

$$= 2222.44 \text{ BDT}$$

$$P_S = \frac{1}{2} \times 66.67 \times (183.33 - 50)$$

$$= 4444.55 \text{ BDT}$$



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$$b) P = 50 + 2Q - 40$$

$$= 10 + 2Q$$

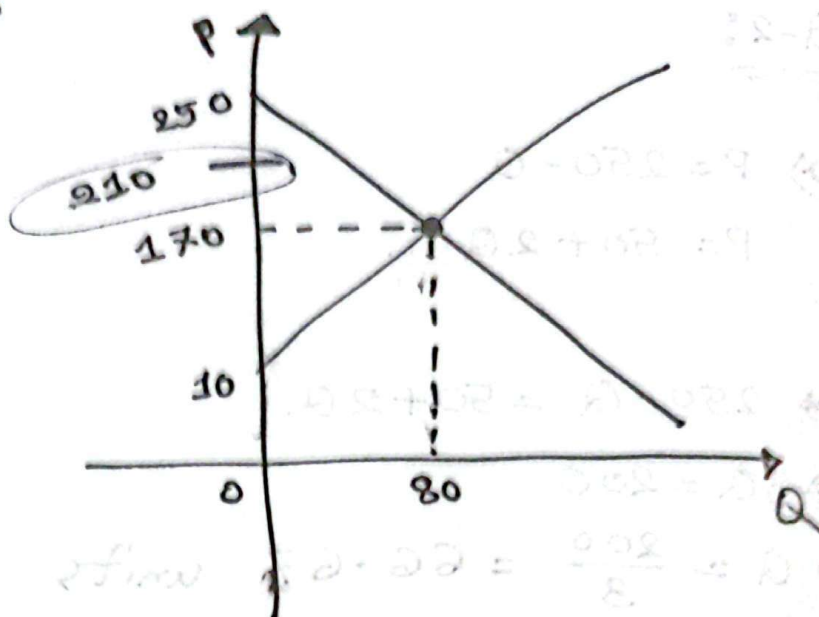
$$P = 250 - Q$$

$$\Rightarrow 250 - Q = 10 + 2Q$$

$$\Rightarrow 3Q = 240$$

$$\therefore Q = 80 \text{ units}$$

$$\therefore P = 170 \text{ BDT}$$



price paid by consumers

$$P = 250 - 80 = 170 \text{ BDT}$$

"received" producers

$$P = 170 + 40 = 210 \text{ BDT}$$

$$c) C_s' = \frac{1}{2} \times 80 \times (250 - 170)$$

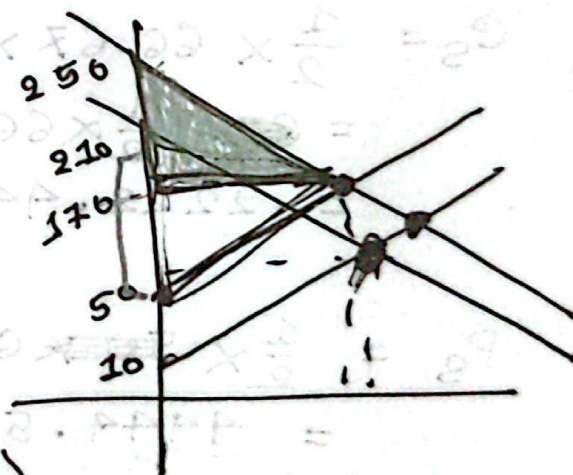
$$= \frac{1}{2} \times 80 \times 80$$

$$= 3200 \text{ BDT}$$

$$P_s' = \frac{1}{2} \times 80 \times (210 - 50)$$

$$= \frac{1}{2} \times 80 \times 160$$

$$= 6400 \text{ BDT}$$



Date

d)

cost x unit

$$\begin{aligned}\text{Subsidy cost to govt.} &= 40 \times 80 \\ &= 3200 \text{ BDT}\end{aligned}$$

dead weight loss due to the subsidy

$$= \frac{1}{2} \times (\text{change in quantity}) \times (\text{subsidy amount})$$

$$= \frac{1}{2} \times (80 - 66.67) \times 40$$

$$= 266.67 \text{ BDT}$$

Date

Elasticity ——— $\left\{ \begin{array}{l} \text{Elastic} \\ \text{Inelastic} \\ \text{Unit Elastic} \end{array} \right.$
(Sensitivity)

$$\text{Elasticity} = \frac{\% \text{ change in effect}}{\% \text{ change in cause}}$$

PED → price elasticity demand

$$E_d = \frac{\% \Delta Q_d}{\% \Delta P}$$

→ High ($|E_d| > 1$) [Elastic]

→ Low ($|E_d| < 1$) [Inelastic]

→ Unit ($|E_d| = 1$) [Unit Elastic]

Normal vs midpoint method

$$\frac{50-40}{40} \times 100\% = 25\%$$

$$\frac{40-50}{50} \times 100\% = -20\%$$

$$\frac{50-40}{\left(\frac{50+40}{2}\right)} = \frac{50-40}{45} = 22$$

$$\frac{40-50}{\left(\frac{50+40}{2}\right)} = -22$$

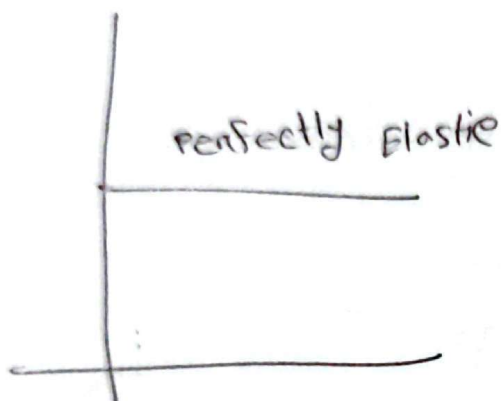
RANGS PHARMA

$$\left(\frac{\frac{40}{120} \times 100}{-\frac{50}{175} \times 100} \right)$$

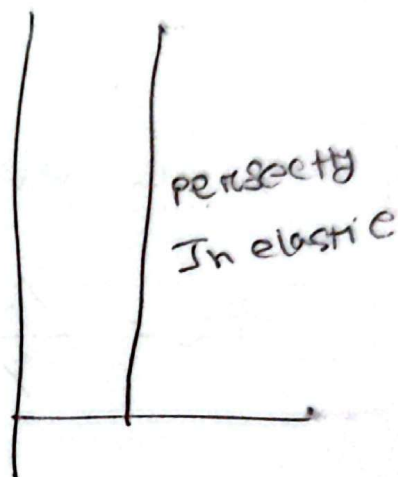
= -1.16

Price is less

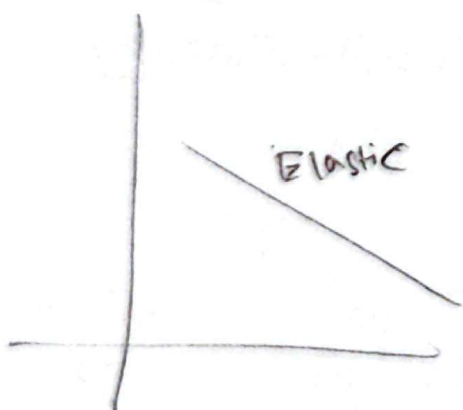
So, Elastic



Fixed in price



Even if the price is high I need that



Price ↑
Demand ↓

Date

PES $\rightarrow$ Price Elasticity of Supply

$$PES = \left(\frac{\% \Delta Q_s}{\% \Delta P} \right)$$

XED $\rightarrow$ Cross price Elasticity

$$E_{xy} = \left(\frac{\% \Delta Q_s}{\% \Delta P_y} \right)$$

substitutes : $E_{xy} > 0$

complements : $E_{xy} < 0$

YED $\rightarrow$ Income Elasticity of Demand

$$E_y = \left(\frac{\% \Delta Q_d}{\% \Delta I} \right)$$

Normal goods : $E_y > 0$

Inferior " : $E_y < 0$

Luxury " : $E_y > 1$

RANGS PHARMA

0.82 $\rightarrow$ substitute

(Price increases
and other products
demands increases)

22 $\rightarrow$ luxury good

(Income $\uparrow$
Demand $\uparrow$)

more Inelastic $\rightarrow$ will pay more tax



what your results
says?

Date

$$\# \quad Q_d = 100 - 10P$$

$$Q_s = 20 + 10P$$

$$Q_d = 100 - 0 = 100$$

$$Q_s = 20 + 0 = 20$$

$$\Rightarrow 0 = 100 - 10P$$

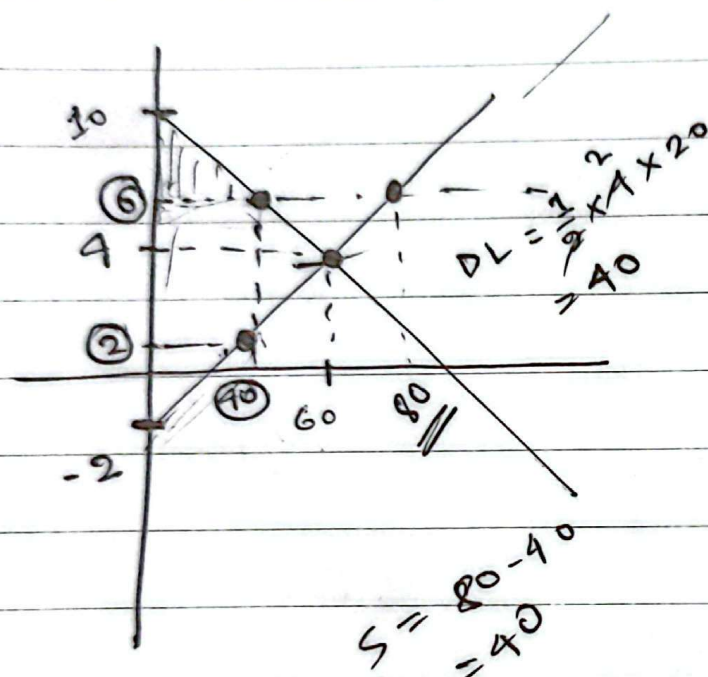
$$\Rightarrow 10P = 100$$

$$\therefore P = 10$$

$$0 = 20 + 10P$$

$$\Rightarrow 10P = -20$$

$$\therefore P = -2$$



$$100 - 10P = 20 + 10P$$

$$\Rightarrow 20P = 80$$

$$\therefore P = 4$$

$$Q_d = 100 - 10 \times 4$$

$$= 60$$

$$C_s = \frac{1}{2} \times (10 - 4) \times 60$$

$$= \frac{1}{2} \times 6 \times 60$$

$$= 3 \times 60$$

$$= 180$$

$$C_s = \frac{1}{2} \times (10 - 6) \times 40$$

$$= \frac{1}{2} \times 4 \times 40$$

$$= 80$$

$$Q_d = 100 - 10 \times 6$$

$$= 100 - 60$$

$$= 40$$

$$P_s = \frac{1}{2} \times (4 + 2) \times 60$$

$$= \frac{1}{2} \times 6 \times 60$$

$$= 3 \times 60$$

$$= 180$$

$$P_s = \left[\frac{1}{2} \times (2 + 2) \times 40 \right]$$

$$+ [40 \times 4]$$

$$Q_s = 20 + 10P$$

$$\Rightarrow 40 = 20 + 10P$$

$$\Rightarrow 10P = 20$$

$$\therefore P = 2$$

RANGS PHARMA

$$= \frac{1}{2} \times 4 \times 40 + 160 = 240$$

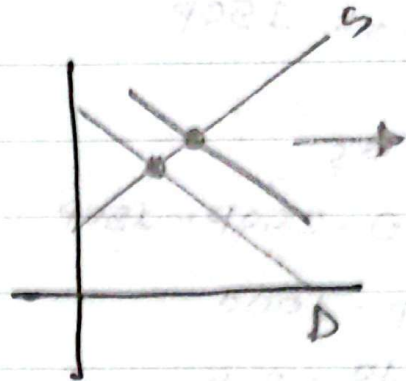
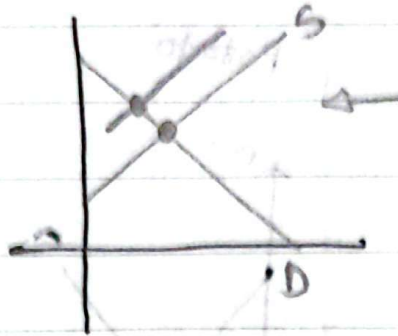
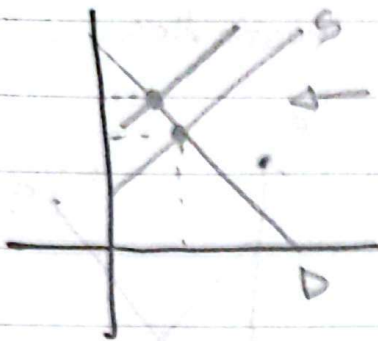
Date

[Practice - MID]

1) Sugar cane

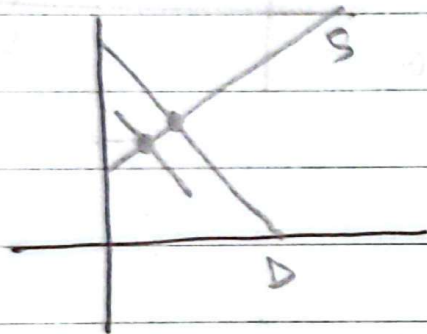
Rum

Whiskey

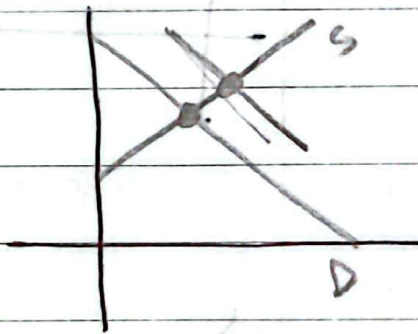


2) hotdogs

butter bun



$P \downarrow Q \downarrow$



$P \uparrow Q \uparrow$

3) $Q_s = 5 + 4P$

$Q_d = 9 + 3P$

$Q_s = Q_d$

$\Rightarrow 5 + 4P = 9 + 3P$

$\Rightarrow P = 4$

(Ans)

$Q = 5 + 4 \times 4$

$= 21$

$\rightarrow P = 6,$

$Q_s = 5 + 4 \times 6 = 29$

$Q_d = 9 + 3 \times 6 = 27$

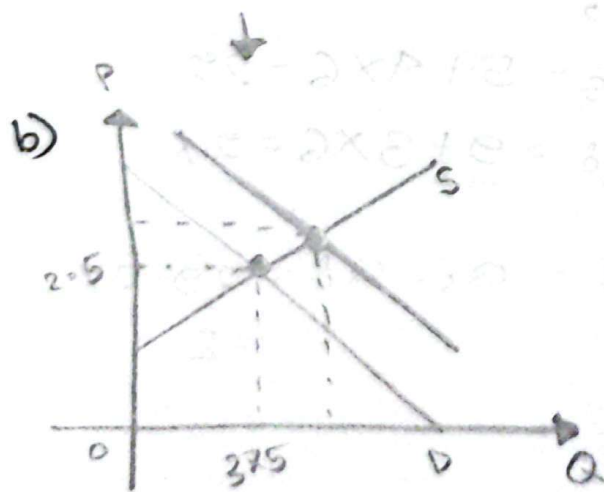
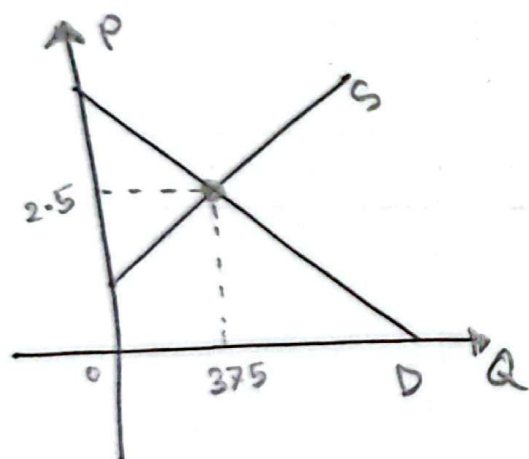
Surplus $= Q_s - Q_d = 29 - 27 = 2$

RANGS PHARMA

1) $Q_d = 1000 - 250P$
 a) $Q_s = 150P$

$\Rightarrow Q_d = Q_s$
 $\Rightarrow 1000 - 250P = 150P$
 $\Rightarrow 400P = 1000$
 $\Rightarrow P = \frac{10}{4} = 2.5$

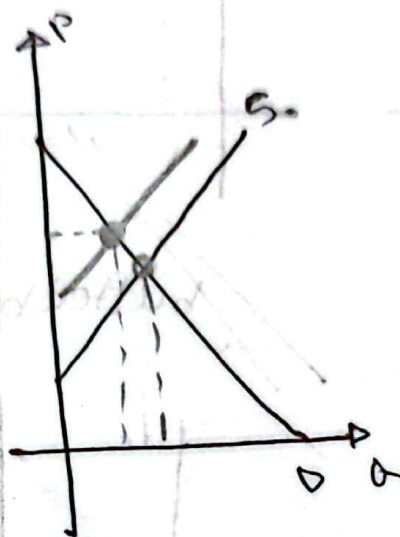
$\therefore Q = 150 \times 2.5$
 $= 375$



$P \uparrow Q \uparrow$

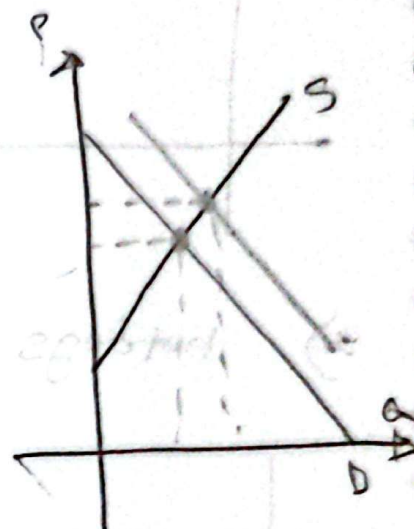
c) ✓

Potato



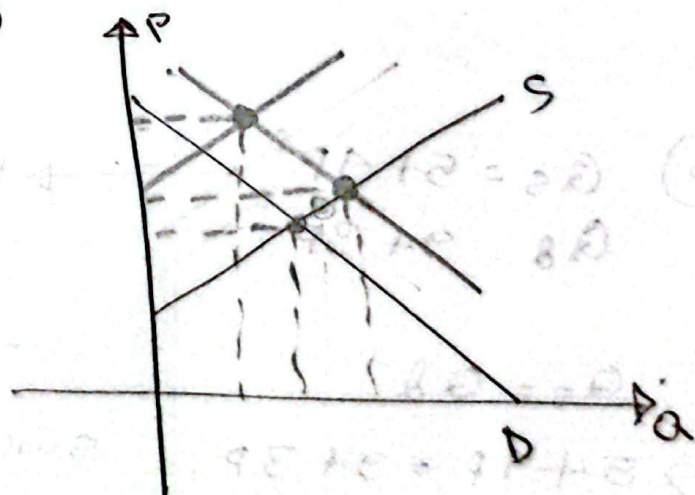
$P \uparrow Q \downarrow$

Pesticides



$P \uparrow Q \uparrow$

d)



$P \uparrow Q \downarrow$

Date

[PPF]

1) company	vanilla	oe _v	chocolate	oe _c
Delight	60	0.5	30	2
Frosty	40	2.25	90	0.44
Snowy	70	0.71	50	1.4

a) 20 liters vanilla Ice cream

b) Snowy $\rightarrow$ absolute adv. $\rightarrow$ vanilla

c) Delight $\rightarrow$ comparative adv. (OE based) $\rightarrow$ vanilla

d) Frosty $\rightarrow$ " " (") $\rightarrow$ chocolate

2) country	Food	oe _F	clothes	oe _c
A	25	3	75	0.33
B	55	0.82	45	1.22
	55-25=30	(1)	75-45=30	(1)
B	35	1.86	65	0.54
	65	0.62	40	1.625
	65-35=30	(0.83)	65-40=25	(1.2)

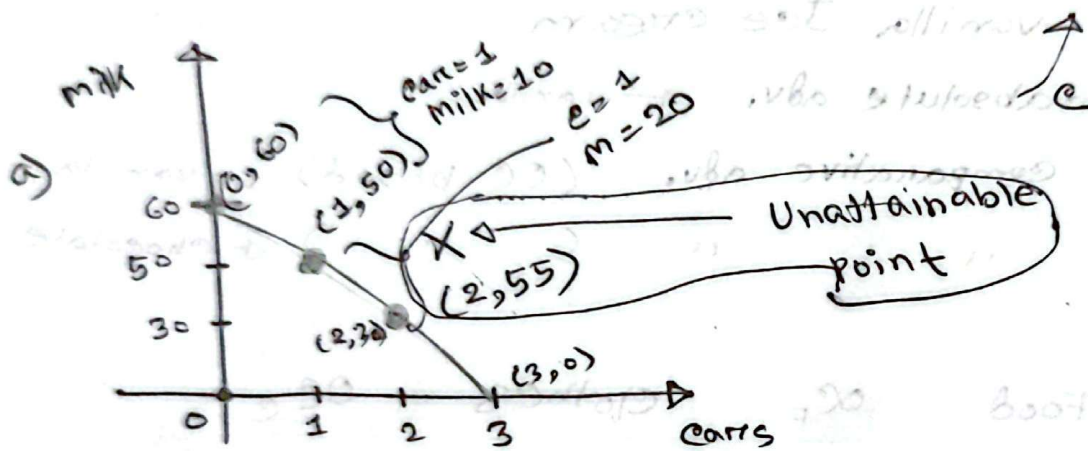
a) Food $\rightarrow$ comparative adv. $\rightarrow$ B

clothes $\rightarrow$ " $\rightarrow$ A

RANGS PHARMA

3) c)

Points	cars	oc _c	milk	oc _m
A	0		60	0
B	1	50	50	0.02
C	2	15	30	0.07
D	3	0	0	



b) ?

	Car	oc _c	milk
A → B	1	10	60 - 50 = 10
B → C	1	20	50 - 30 = 20
C → D	1	30	30 - 0 = 30

4) a)

	Juice	oc _j	wraps	oc _w
Anna	10	(1)	10	1
Leo	2.5	6	15	(0.17)
	10 - 6 = 4	5.5	10 + 12 = 22	0.18
	2.5 + 6 = 8.5	0.35	15 - 12 = 3	2.83

Date

11/11/2021

5)	Shirt	UP _{sh}	Scarf	OP _{sc}
Marcia	4	2	2	0.5
John	3	3	1	0.3

a) abs. adv. $\rightarrow$ scarf $\rightarrow$ John

b) n $\rightarrow$ shirt $\rightarrow$ John

c)

comp. adv. $\rightarrow$ shirt $\rightarrow$ Marcia

scarf $\rightarrow$ John

[Elasticity]

$$1) \quad PED = \frac{\frac{400 - 500}{400 + 500} \times 100}{\frac{1.50 - 1}{1.50 + 1} \times 100}$$

$$= -0.56 < 1 ; \text{Inelastic}$$

b) Total Revenue,
 before = $1 \times 500 = 500$ pounds
 after = $1.5 \times 400 = 600$ pounds

$P \uparrow R \uparrow$; so demand is inelastic.

$$c) \quad YED = \frac{\frac{150 - 100}{125} \times 100}{\frac{3300 - 3000}{3150} \times 100}$$

$$= 1.24 ; \text{Elastic}$$

$$> 1 ; \nearrow$$

so, luxury goods

$$ED = \frac{\% \Delta Q}{\% \Delta P}$$

$|ED| > 1$; Elastic ^{luxury}

$|ED| < 1$; Inelastic ^{necessity}

$|ED| = 1$; Unit Elastic

Normal goods : $E_y > 0$

Inferior goods : $E_y < 0$

Luxury goods : $E_y > 1$

XED

substitutes : $E_{xy} > 0$

complements : $E_{xy} < 0$

Date

[13/09/2023 & 30/09/2023]

3)

$$PES = \frac{\% \Delta Q}{\% \Delta P} = \frac{\frac{13-10}{11.5} \times 100}{\frac{30-20}{45} \times 100} = 0.65 < 1, \text{ Somewhat Inelastic}$$

a)

b) labour / time / space



making it difficult to
increase supply quickly

$PES < 1$: ~~Elastic~~
Inelastic

[Welfare & Efficiency]

1) ~~Q~~ $P = 32 - 8Q \triangleleft D$
 $P = 12 + 2Q \triangleleft S$

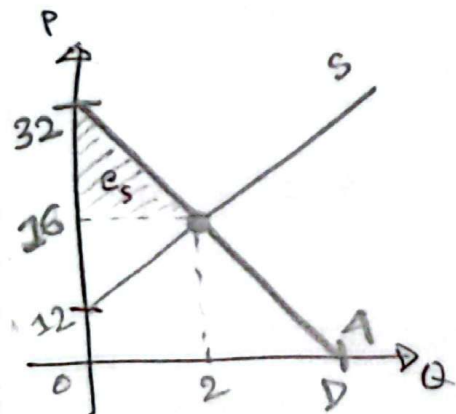
$$\Rightarrow 32 - 8Q = 12 + 2Q$$

$$\Rightarrow 20 = 10Q$$

$$\therefore Q = 2$$

$$P = 12 + 2 \times 2 \quad | \quad P = 16$$

$$= 16 \quad | \quad Q = 2$$



$$P = 32 - 8 \times 0$$

$$= 32$$

$$Q = 32 - 8Q$$

$$\Rightarrow 8Q = 32$$

$$\therefore Q = 4$$
~~$$P = 16 - 32 - 8Q$$~~
~~$$Q = 2$$~~

$$P = 12 + 2 \cdot 0$$

$$= 12$$

$$CS = \frac{1}{2} \times (32 - 16) \times 2$$

$$= 16$$

$$PS = \frac{1}{2} \times (16 - 12) \times 2$$

$$= 4$$

2)

Quantity Demand

= quantity supplied

No deadweight loss

maximized total surplus

c)

$$\text{Total Surplus} = 16 + 4$$

$$= 20$$

Date

2)

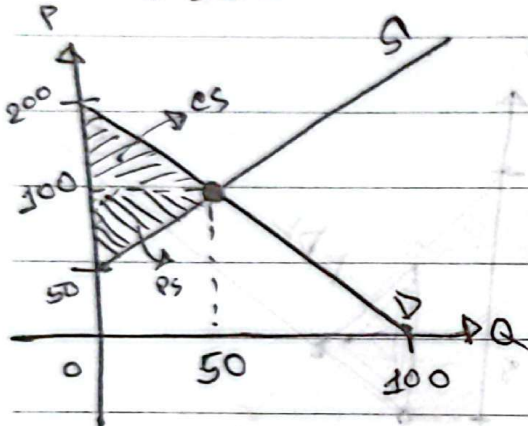
$$a) P = 200 - 2Q \quad \leftarrow D$$

$$P = 50 + Q \quad \leftarrow S$$

$$\Rightarrow 200 - 2Q = 50 + Q$$

$$\Rightarrow Q = 50$$

$$\therefore P = 50 + 50 \\ = 100$$



b)

$$Q=0, P=200$$

$$Q=0, P=50$$

$$C_s = \frac{1}{2} \times 100 \times 50 \\ = 2500$$

$$P_s = \frac{1}{2} \times 50 \times 50 \\ = 1250$$

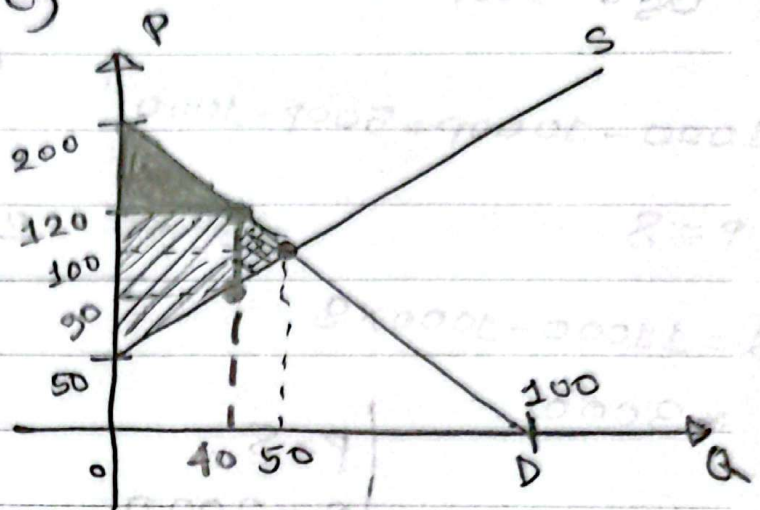
$$P=0, \quad$$

$$0 = 200 - 2Q$$

$$\Rightarrow 2Q = 200$$

$$\therefore Q = 100$$

c)



$$P=120,$$

$$120 = 200 - 2Q$$

$$\Rightarrow Q = 40$$

$$C_s' = \frac{1}{2} \times (200 - 120) \times 40 \\ = 1600$$

$$P_s' = \frac{1}{2} \times (120 - 50) \times 40 + \frac{1}{2} \times (120 - 90) \times 10 \\ = 800 + 1200 \\ = 2000$$

$$Q=40,$$

$$P = 50 + 40 \\ = 90$$

$$\therefore C_s' = 1600$$

$$P_s' = 2000$$

$$DL = \frac{1}{2} \times (100 - 90) \times (50 - 40) \\ = 150$$

RANGS PHARMA

3)

$$Q_d = 11000 - 1000P$$

$$Q_s = 500P - 1000$$

$$11000 - 1000P = 500P - 1000$$

$$\therefore P = 8$$

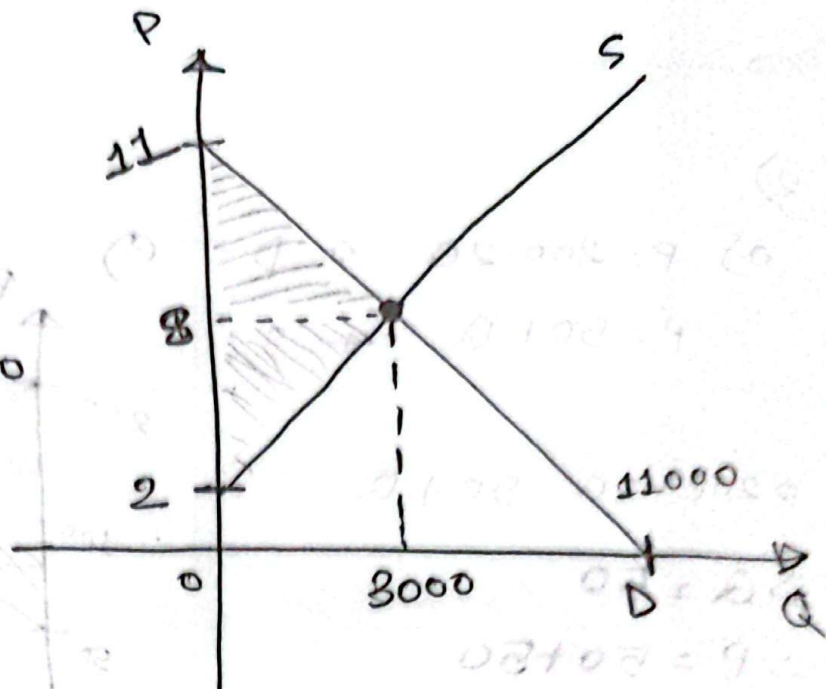
$$Q = 11000 - 1000 \times 8$$

$$= 3000$$

$$P = 0, Q = 11000$$

$$Q = 0, P = 11$$

$$Q = 0, P = 2$$



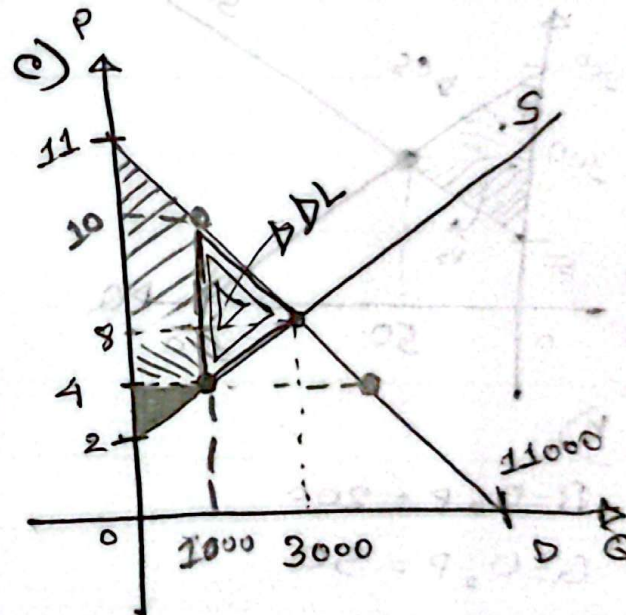
$$\left| \begin{array}{l} P = 8 \\ Q = 3000 \end{array} \right.$$

$$C_s = \frac{1}{2} \times (11 - 8) \times 3000$$

$$= 4500$$

$$P_s = \frac{1}{2} \times (8 - 2) \times 3000$$

$$= 9000$$



$$P = 4, Q = \frac{500 \times 4 - 1000}{1000 - 500} = 1000$$

$$\Rightarrow Q = 1000$$

$$C_s' = \frac{1}{2} \times (11 - 4) \times 1000 + \frac{1}{2} \times (4 - 2) \times 1000$$

$$= 6500$$

$$P_s' = \frac{1}{2} \times (4 - 2) \times 1000$$

$$= 1000$$

$$\therefore C_s' = 6500$$

$$P_s' = 1000$$

$$DL = \frac{1}{2} \times (11 - 4) \times 2000$$

$$= 6000$$

(Ans)

Date

a)

$$P = 120 - 3Q \leftarrow D$$

$$P = 30 + 2Q \leftarrow S$$

$$120 - 3Q = 30 + 2Q$$

$$\therefore Q = 18$$

$$\therefore P = 66$$

$$P = 120 - 3Q \leftarrow D$$

$$P = 30 + 2Q + 15 \leftarrow S$$

$$= 45 + 2Q$$

$$120 - 3Q = 30 + 2Q + 15$$

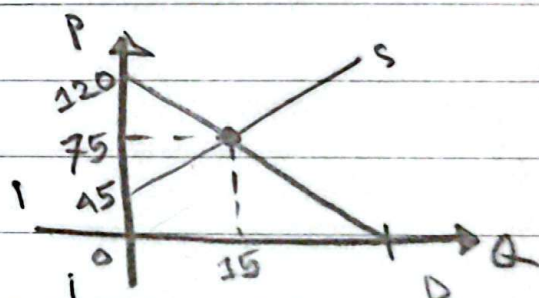
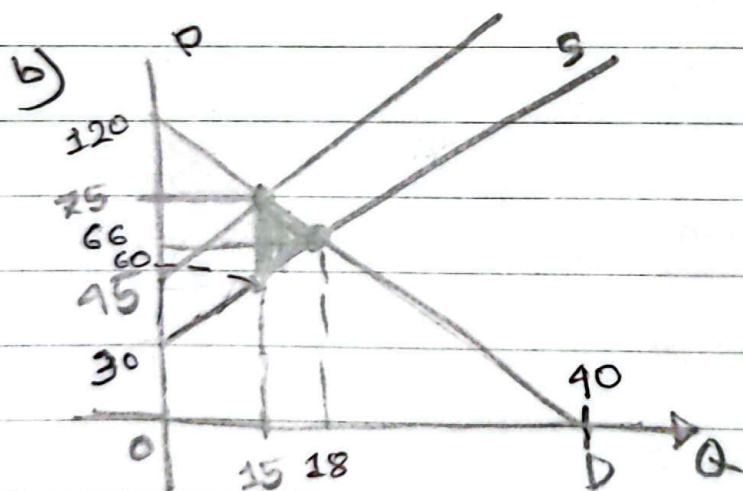
$$\therefore Q' = 15$$

$$\therefore P' = 75$$

Price buyers pay = \$75

Price sellers receive = \$75 - \$15 = \$60

Quantity sold = 15



$$Q = 15,$$

$$P = 45 + 2 \times 15$$

$$= 75$$

$$C_s = \frac{1}{2} \times (120 - 66) \times 18$$

$$= 486$$

$$P_s = \frac{1}{2} \times (66 - 30) \times 18$$

$$= 324$$

$$C_s' = \frac{1}{2} \times (120 - 75) \times 15$$

$$= 337.5$$

$$P_s' = \frac{1}{2} \times (75 - 45) \times 15$$

$$= 225$$

e)

$$DL = \frac{1}{2} \times (75 - 60) \times (18 - 15)$$

$$= 22.5$$

$$P = 30 + 2 \times 15$$

$$= 60$$

RANGS PHARMA

c)

Tax Revenue = Tax per unit $\times$ Quantity sold

$$= 15 \times 15 = 225$$

$$= 225$$

$$25 = 0$$

$$25 = 9$$

$$25 = 0$$

$$25 = 0$$

$$25 = 9$$

$$25 = 13 + 12$$

$$25 =$$

(9)

$$25 = 10$$

$$25 = 10$$

$$25 = 10$$

$$25 = 10$$

$$25 = 10$$

Date

Types of Cost:

1) Fixed cost

Rent Fee
→ per unit price is always same

2) variable cost

→ per unit price change

3) Total cost

4) Marginal cost

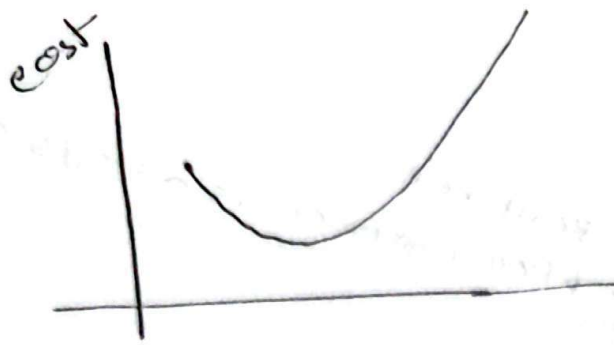
price

→ how much will be change
in respect to previous
unit

$$= \frac{\text{change in total cost}}{\text{change in output}}$$

output	variable cost (cumulative)	fixed cost	Total cost	Marginal cost
0	0	10	10	0
1	10	10	20	$\frac{20-10}{1-0} = 10$
2	17	10	27	$\frac{27-20}{2-1} = 7$
3	25	10	35	$\frac{35-27}{3-2} = 8$
4	40	10	50	$\frac{50-35}{4-3} = 15$
5	60	10	70	$\frac{70-50}{5-4} = 20$
6	110	10	120	$\frac{120-70}{6-5} = 50$

RANGS PHARMA



while, less productive.

I need more time to do the same thing

later, that's why

(time cost) is rising.

Short term $\rightarrow$ Marginal cost $= \frac{\Delta c}{\Delta Q}$

Long term $\rightarrow$ Avg. ^{Total} cost

Avg. variable cost (AVC) $= \frac{\text{Variable cost}}{\text{Quantity}}$

Avg. Fixed cost (AFC) $= \frac{\text{Fixed cost}}{\text{Quantity}}$

Avg. Total cost (ATC) $= \frac{\text{Total cost}}{\text{Quantity}}$

Marginal Revenue $= \frac{\text{change in total revenue}}{\text{change in quantity}}$

Profit $= \text{Total Revenue} - \text{Total cost}$

Date

• If,

Price < AVC

$P < \text{Avg. var. cost}$ [shut down the business]

in the long run

but in the short term

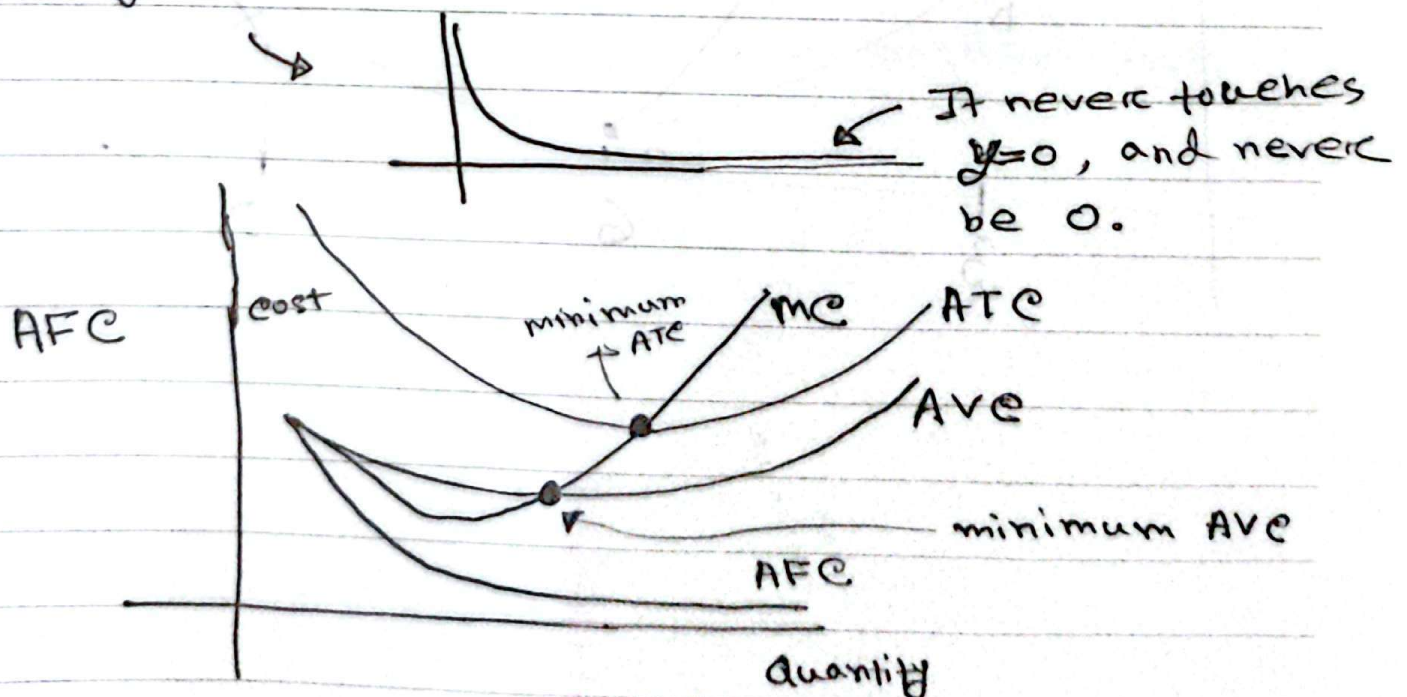
It can make sense

while I wait for

a chance of a change.

• "Profit maximizing" is when $[MR = MC]$

✓ Asymptote



RANGS PHARMA

Date

[10/11/2019]

Increasing returns
to scale

- Specialization
- Natural monopoly cost
- Network effects

Decreasing

- Co ordination
- Communication
- overcrowding

Constants

- ✓ Replication (Bakery)
- ✓ Perfectly divisible inputs

✓ No bottlenecks

[market]

1) perfect competition

2) monopoly



1) many buyers

2) Homogenous products

3) Perfect information

4) No barriers to entry/exit

5) perfect mobility of factors of production

6) Zero transaction costs

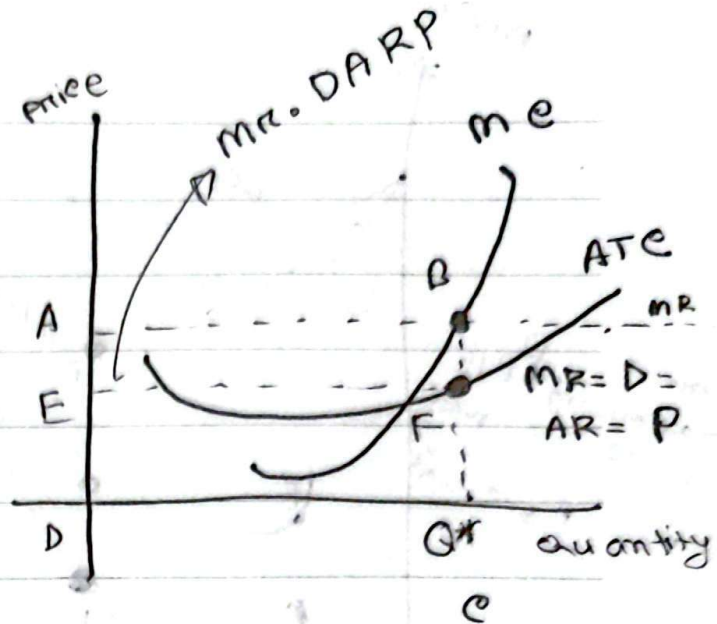
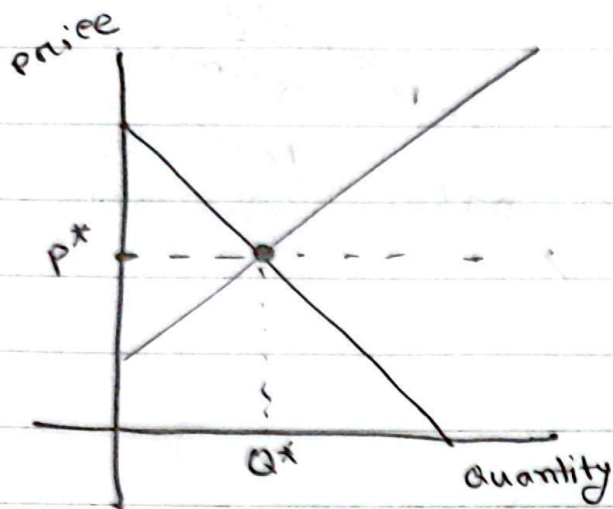
7) Profit maximization

8) No externalities



we compare with these things to know what is the change.

Date



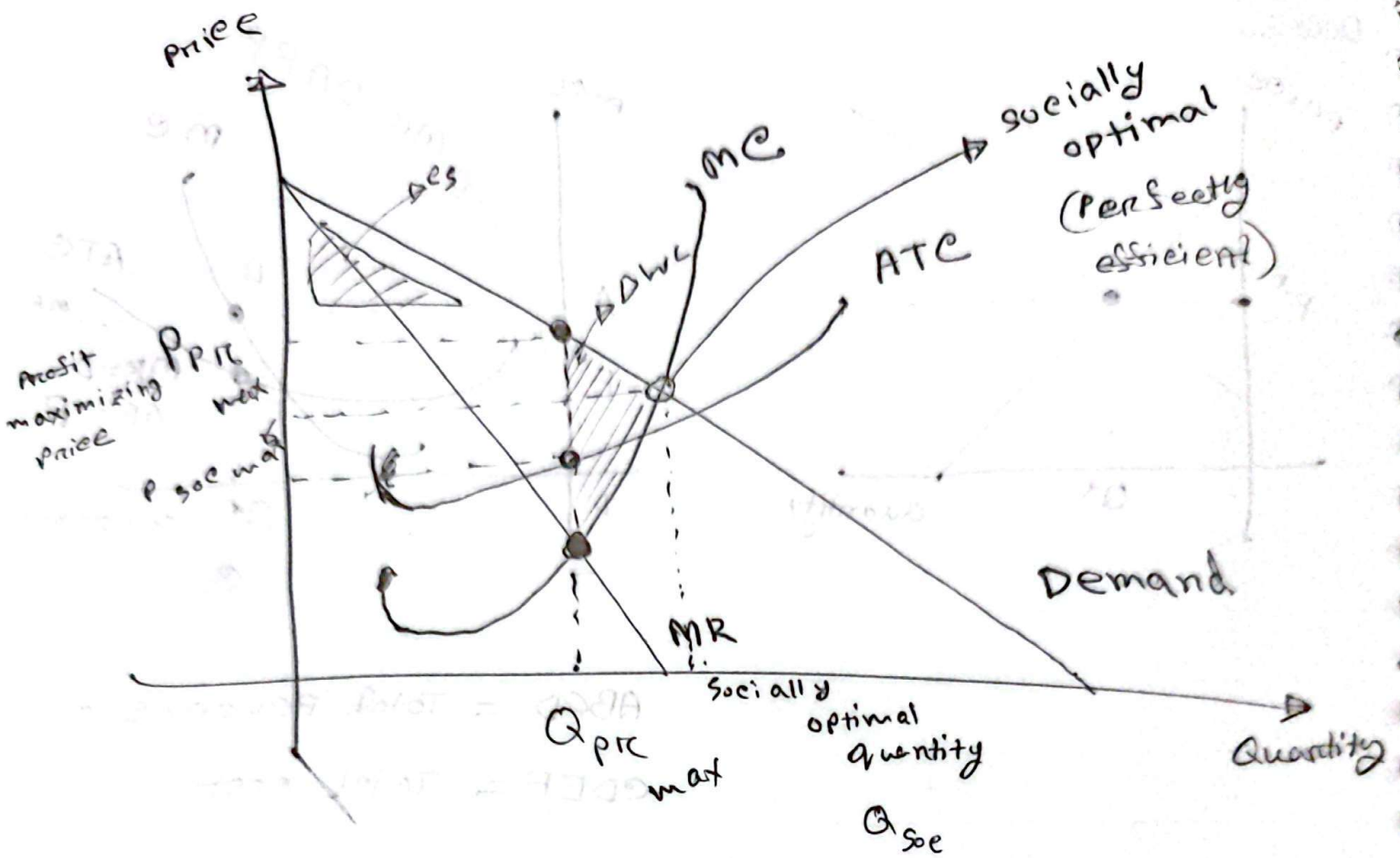
$ABCD = \text{Total Revenue}$

$CDEF = \text{Total cost}$

monopoly:

- 1) Single seller (No competitors)
- 2) No close substitutes
- 3) High barriers to entry
- 4) Price maker
- 5) Profit maximization
- 6) Downward sloping demand curve
- 7) No price discrimination

RANGS PHARMA



$$DWL = \frac{1}{2} \times (Q_{SOC} - Q_{PR\ max}) \times (P_{PR\ max} - P_{SOC\ max})$$

Date

④

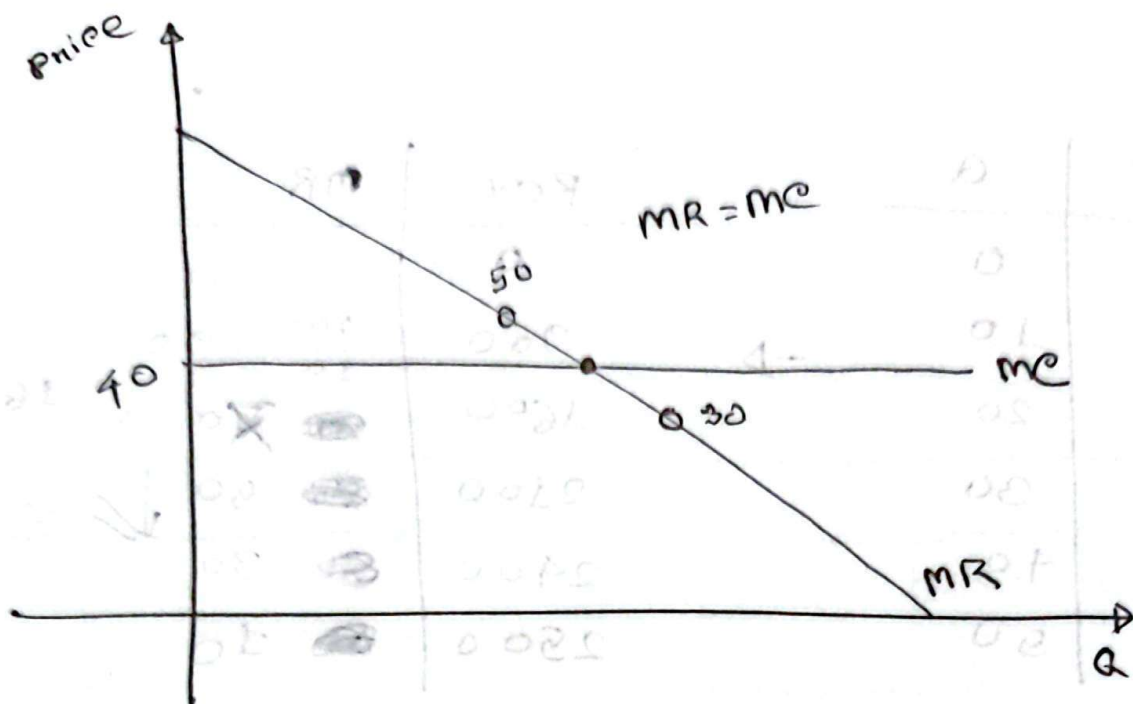
P	Q	Rev	MR
100	0	0	
90	10	900	$\frac{900}{10} = 90$
80	20	1600	80 $= \frac{1600 - 900}{10}$
<u>70</u>	30	2100	70 50
60	40	2400	60 30
50	50	2500	50 10

marginal cost is constant at 40.

Avg " " also 40 (no fixed cost)

- Find a monopoly price and quantity.
- Under perfect competition, what are the price and quantity?
- compute the monopolist's profit and compare with profit under perfect competition.
- compute the deadweight loss (DWL) from monopoly.

RANGS PHARMA



a)

Profit maximizing price = 70
 Quantity = 30

monopoly Price

monopoly Quantity

perfect competition

$P=MC$

b)

P	Q	Rev	MR
40	0	0	0
40	10	400	40
40	20	800	40
40	30	1200	40
40	40	1600	40
40	50	2000	40

max quantity

✓ Firms sell at cost

✓ zero profit

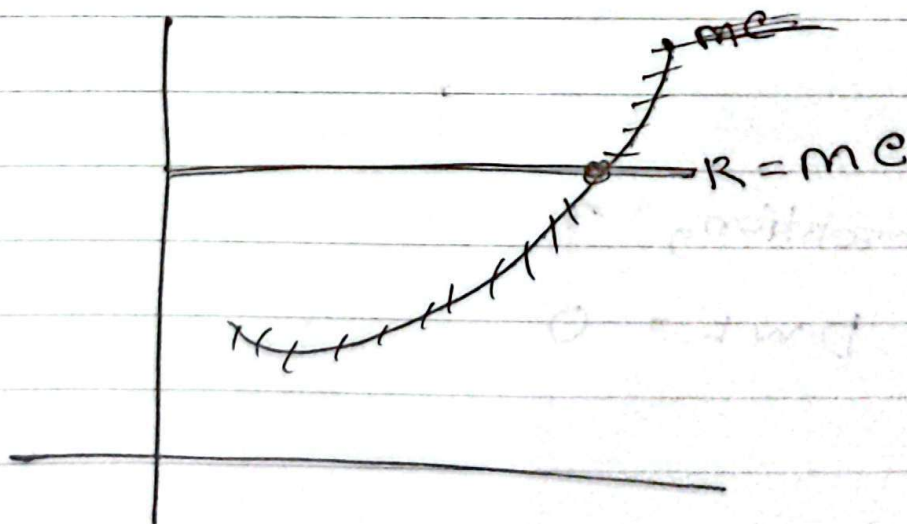
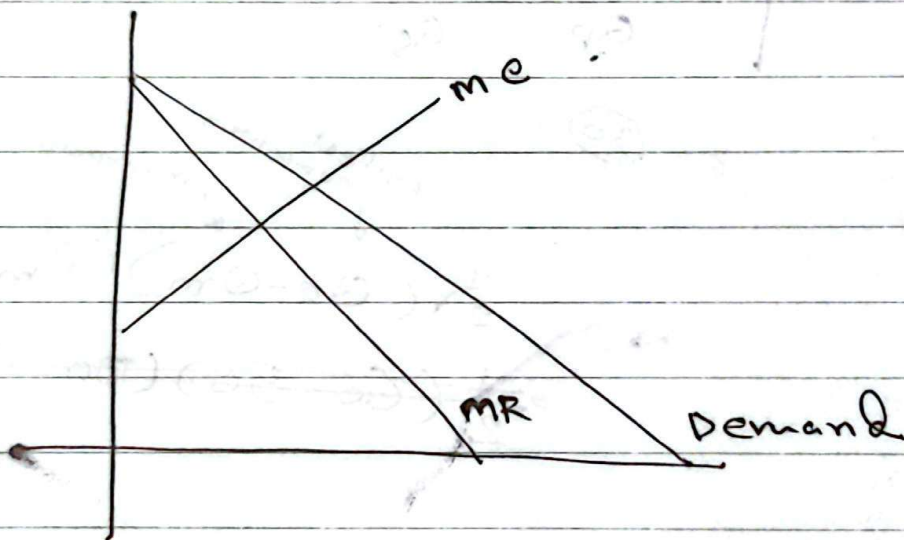
Date

c) monopolist,

$$\begin{aligned}\text{profit} &= (P - AC) \times Q_m \\ &= (30 - 40) \times 30 \\ &= 900\end{aligned}$$

Perfect competition,

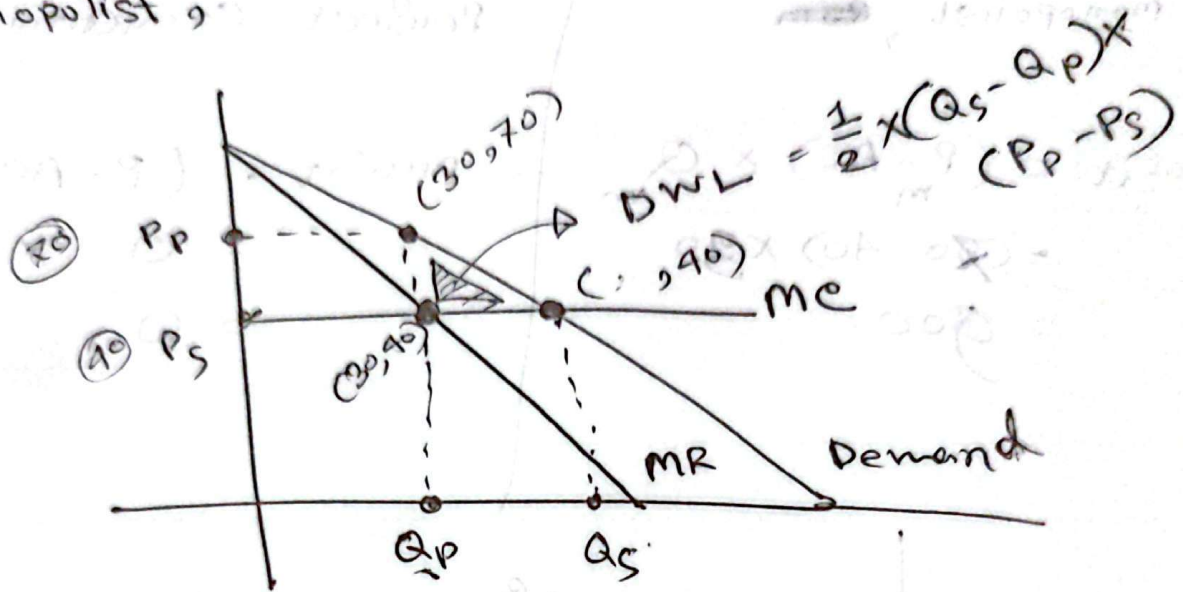
$$\begin{aligned}\text{profit} &= (P - AC) \times Q \\ &= 0 \times Q \\ &= 0\end{aligned}$$



RANGS PHARMA

2)

monopolist 9



30

perfect competition

monopoly

$$\frac{1}{2} \times (Q_e - Q_m) (P_m - P_e)$$

$$= \frac{1}{2} \times (60 - 50) (50 - 40)$$

Perfect competition

$$DWL = 0$$

Date

(2)

<u>Q</u>	<u>TC</u>	<u>VC</u>	<u>P</u>	<u>TFC</u>	<u>APC</u>	<u>AVC</u>	<u>MC</u>	<u>TR</u> (P×Q)
0	50	0	60	50	—	—	—	0
1	70	20	60	50	50	20	20	60
2	90	40	60	50	25	20	20	120
3	120	70	60	50	16.67	23.33	30	180
4	160	110	60	50	12.5	27.5	40	240
5	210	160	60	50	10	32	50	300

<u>Profit</u>	<u>TR</u>	<u>Profit</u>
-50	0	-50
-10	18	-52
30	36	-54
60	54	-66
80	62	-78
90 ← maximum profit	90	-130

Minimum $AVC \approx 18$ at low Q.

Since $P \approx 18 < \min AVC$, shut down

prisoners Dilemma

$$1, 4$$
$$3, 3$$

equilibrium

Nash equilibrium can be multiple.

1, 3

$$3, 3$$

we need to check strict dominance.

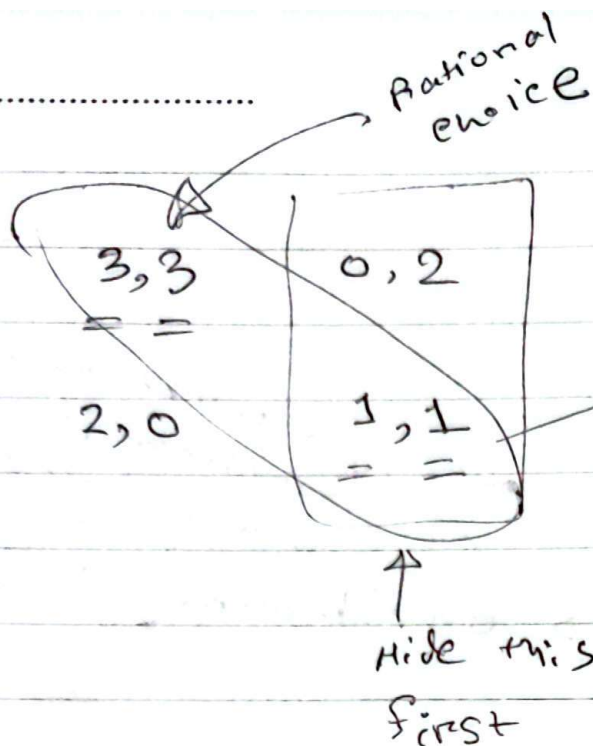
↳ cut down the 3

Nash Equilibrium:

↳ All of the other options are worse except the chosen one.

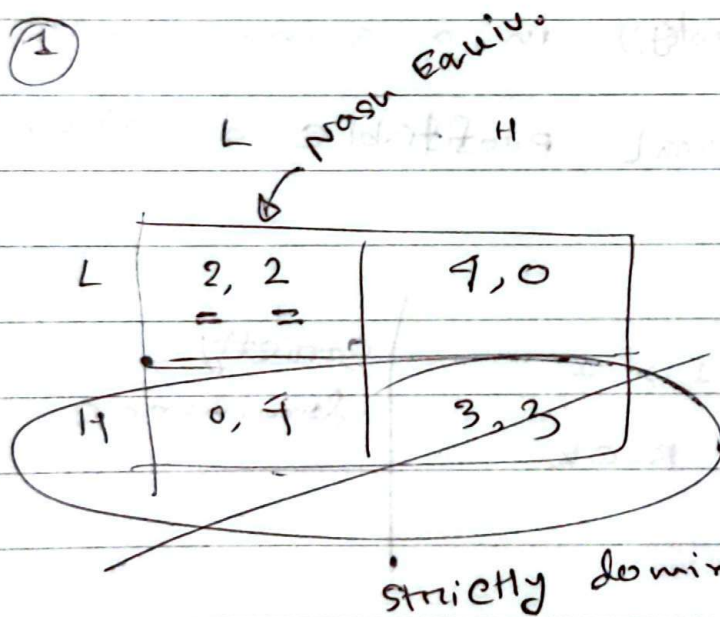
unilateral ~~and~~ profitable deviation,

Date

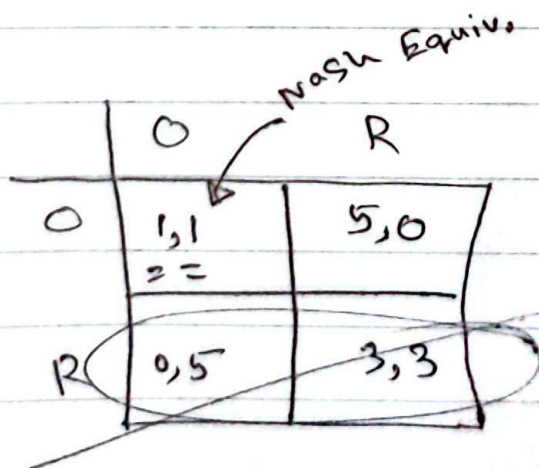


weak dominance can't be neglected

①



②



Tradeoff of the commons

RANGS PHARMA

Payoff matrix:

He (P1)

She (P2)

	R	P
R	1, 1	-1, 5
P	5, -1	1, 1

It will be nash equilibrium if there is no other unilateral profitable deviation

equilibrium

Nash equi. such a strategy in a game where no player has unilateral profitable deviation.

For P1 $1 > 5$ $-1 > 1$
 so we will not play Rock

Strictly dominance

Date

3, 4	5, 6
1, 2	7, 9

→ Nash Equiv.

↓
What does that mean?

[Practice]

(Game Theory)

P-2

Quiet

Rat

Quiet

-1, -1

-12, 0

P-1

Rat

0, -12

-8, -8

left

center

Right

up

13, 3

1, 4

7, 3

middle

4, 1

3, 3

6, 2

Down

-1, 9

2, 8

8, -1

Nash Equilibrium

strictly

dominates right

Date

1)

	Low	High
Low	(<u>2</u> , <u>2</u>)	(4, 0)
High	(0, 4)	(3, 3)

Nash equilibrium

Undercutting to a low price always beats a high price regardless of the rival's choice, so both end up choosing L.

2)

	O	R
O	(<u>1</u> , <u>1</u>)	(5, 0)
R	(0, 5)	(3, 3)

Nash equilibrium

- Individual rational behavior leads to (4, 1)
- collective cooperation would yield (3, 3) - better for both - but is unstable since any fisher can gain by deviating to O (from 3 to 5)

3)

	R	W	T
R	(0, 0)	(3, 1)	(<u>2</u> , <u>2</u>)
W	(1, 3)	(2, 2)	(1, 4)
T	(<u>2</u> , <u>2</u>)	(4, 1)	(1, 1)

NE

4)

	A	P	O
A	$(-5, -5)$	$(6, 1)$	$(4, 0)$
P	$(1, 6)$	$(2, 2)$	$(3, 0)$
O	$(0, 4)$	$(0, 3)$	$(0, 0)$

column O is strictly dominated by column P

NE

Date [CS, PS, Tax, DWL]

$$\textcircled{1} \quad P_d = 120 - 1.5Q$$

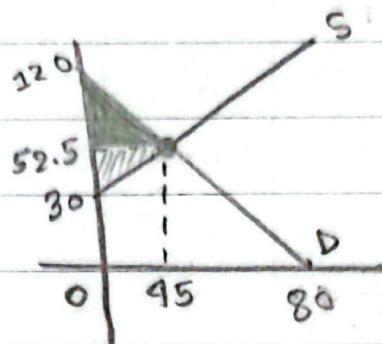
$$P_s = 30 + 0.5Q$$

$$a) \quad 120 - 1.5Q = 30 + 0.5Q$$

$$\Rightarrow 2Q = 90$$

$$\therefore Q = 45$$

$$\therefore P = 30 + 0.5 \times 45 = 52.5$$



$$CS = \frac{1}{2} \times (120 - 52.5) \times 45$$

$$= 1518.75$$

$$PS = \frac{1}{2} \times (52.5 - 30) \times 45$$

$$= 506.25$$

$$b) \quad P_d = 120 - 1.5Q$$

$$P_s = 30 + 0.5Q$$

Government imposes a BDT 20 per litre tax on producers.

$$S_0, \quad P_s = 30 + 20 + 0.5Q$$

$$= 50 + 0.5Q$$

$$S_0, \quad 120 - 1.5Q = 50 + 0.5Q$$

$$\Rightarrow 2Q = 70$$

$$\therefore Q = 35$$

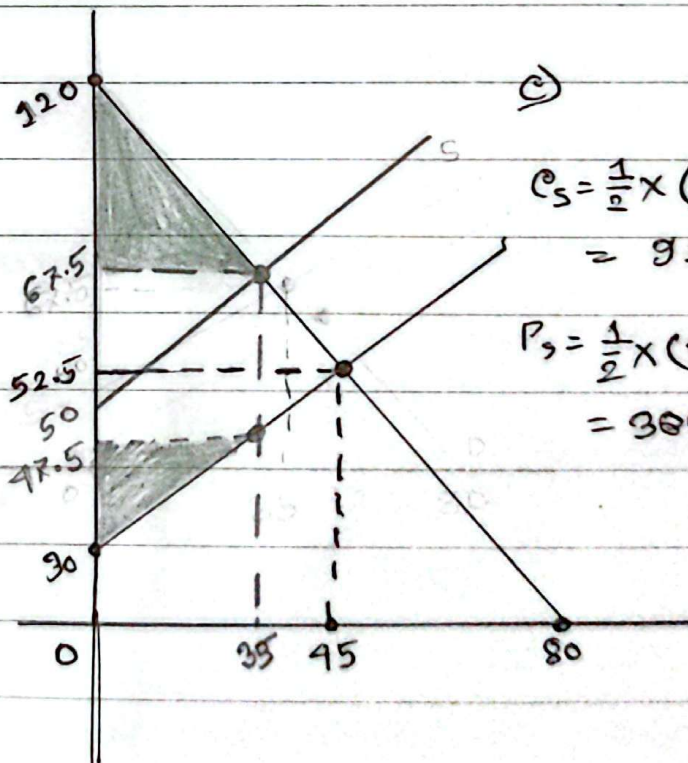
$$\therefore P' = 50 + 0.5 \times 35$$

$$= 67.5$$

after tax,

$$P'' = 67.5 - 20$$

$$= 47.5$$



$$CS = \frac{1}{2} \times (120 - 67.5) \times 35$$

$$= 918.75$$

$$PS = \frac{1}{2} \times (47.5 - 30) \times 35$$

$$= 306.25$$

$$DWL = \frac{1}{2} \times (67.5 - 47.5) \times 10$$

$$= 100 \text{ BDT}$$

2)

Tax Revenue = price $\times$ quantity

$$= 20 \times 35$$

$$= 700$$

RANGS PHARMA

Date 1

L	Q	P	TC	FC	VC	ATC	AVC	TR	MC	MR
0	0	20	120	120	0	—	—	—	—	
1	5	20	160	120	40	32	8	100	8	20
2	9	20	200	120	80	22.22	8.89	180	10	20
3	13	20	260	120	140	20	10.77	260	15	20
4	16	20	320	120	200	20	12.5	320	20	20
5	18	20	400	120	280	22.22	15.56	360	40	20
6	19	20	480	120	360	25.26	18.95	380	80	20

b) perfectly comp. market,

Profit maximizing level, $MR = MC$

$$20 = 20$$

when, $P = 20, Q = 16$

$$\text{Profit} = TR - TC$$

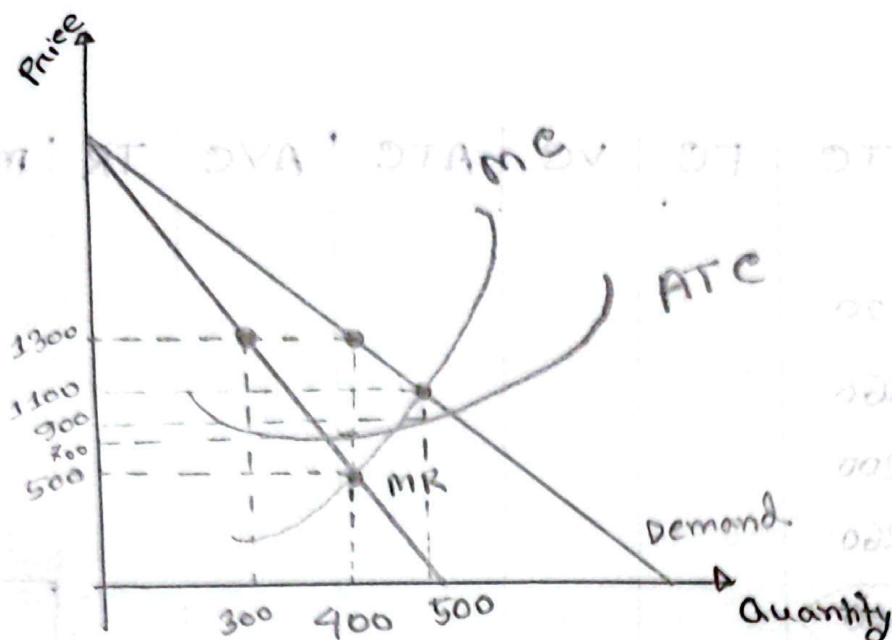
$$= 320 - 320 = 0$$

c) In the short run, even if profit is zero.

Producer should not shut down.

RANGS PHARMA

②



a) $MR = MC$

but, In monopoly, profit maximizing price = ~~1100~~ 1300
and, " " " quantity = ~~500~~ 400

b) In competitive market,

price will be 1100
Quantity " " 500

since,

$Demand = MC$

c) Under monopoly,

$$\begin{aligned} \text{Profit} &= TR - TC \\ &= 520000 - 280000 \\ &= 240000 \end{aligned}$$

$$\begin{aligned} TR &= 1300 \times 400 \\ &= 520000 \\ TC &= 700 \times 400 \\ &= 280000 \end{aligned}$$

$MR = MC$
 ATC
 $P = 1300$
 $Q = 400$
 TC

Date

under perfect comp.,

$$\begin{aligned}\text{Profit} &= \text{TR} - \text{TC} \\ &= 550000 - 450000 \\ &= 100000\end{aligned}$$

$$\begin{aligned}\text{TR} &= 1100 \times 500 \\ &= 550000 \\ \text{TC} &= 900 \times 500 \\ &= 450000\end{aligned}$$

So, $P_{\text{Monopoly}} > P_{\text{perfect comp.}}$

$$\begin{aligned}d) \text{DWL} &= \frac{1}{2} \times (1300 - 500) \times (500 - 400) \\ &= \frac{1}{2} \times 800 \times 100 \\ &= 40000\end{aligned}$$

(Ans)

e) when the market is competitive, then it's more socially efficient. Because, there is

no DWL.

monopolies still exist if there is no close substitute of the product and if there are high barriers to the market.

RANGS PHARMA

$$3) TC = 50 + 2Q^2 + 20Q$$

$$P = 800 - 2Q$$

$$P_s = 200 + Q$$

$$800 - 2Q = 200 + Q$$

$$\Rightarrow 3Q = 600$$

$$\therefore Q = 200$$

$$\therefore P = 200 + 200 = 400$$

$$b) MC = 4Q + 20$$

In perfect comp., $MC = P$

$$MR = 400$$

$$MC = MR = 400$$

$$\Rightarrow 4Q + 20 = 400$$

$$\therefore Q = 95$$

$$\begin{aligned} \text{Profit} &= TR - TC \\ &= 38000 - 20000 \\ &= 18000 \end{aligned}$$

$$\therefore TR = P \times Q = 400 \times 95 = 38000$$

$$TC = 50 + 2(95)^2 + 20 \times 95 = 20000$$

Date

This is a short run equilibrium. Because, there is profit. In the long run equilibrium the profit can be 0.

c) As there's profit, more producers will try to get into the market. At some point profit will become 0.